海上异构无人集群对抗防御决策研究

补充材料

作者：蒋宗宏

本文档收录从学位论文正文中移出的技术细节、提示词模板、实验策略说明与原始数据附表。

2026 年 11 月

文档说明

本补充材料与学位论文《海上异构无人集群对抗防御决策研究》配套使用，收录平台轨迹语义压缩方法、上层指挥决策器提示词模板、覆盖搜索热点方向补偿方法，以及第三至第五章主要实验的原始数据附表。为保证文档能够独立编译，原先指向论文正文的交叉引用已改写为自包含说明；研究内容与数据本身未作扩展。

目录

文档说明 i

附录 A　平台轨迹语义压缩方法 1

附录 B　上层指挥决策器提示词模板 3

B.1　威胁位置推断提示词 3
B.2　态势总结提示词 3
B.3　事实核查提示词 4
B.4　结论封装提示词 4
B.5　资源规划提示词 5
B.6　任务绑定节点规则 5

附录 C　覆盖搜索热点方向补偿方法 6

附录 D　原始数据附表 8

附录 A　平台轨迹语义压缩方法

原始日志逐步记录平台位置与策略，直接输入会随决策间隔和平台数量增长，且难以表达机动或驻留状态。因此，系统提取当前周期样本，经滑动窗口判定、相邻样本合并和事件汇总，生成移动与区域驻留两类语义轨迹段。

设相邻两次上层决策时刻为 t_{k-1}^{H} 和 t_k^{H}。将平台 i 截至 t_k^{H} 的原始日志按采样时刻升序排列，记为

$$\mathcal{T}_{i,k} = \left(\left(t_l, \boldsymbol{p}_i^{t_l}, \omega_i^{t_l}\right)\right)_{l=1}^{L_{i,k}}, \qquad t_1 < t_2 < \cdots < t_{L_{i,k}} \le t_k^{\mathrm{H}}, \tag{A.1}$$

其中，$L_{i,k}$ 为样本数，t_l、$\boldsymbol{p}_i^{t_l}$ 和 $\omega_i^{t_l}$ 分别为时刻、位置和任务标签。本周期仅输出 $t_l \in (t_{k-1}^{\mathrm{H}}, t_k^{\mathrm{H}}]$ 的样本；长度不超过 W_{win} 的判定窗口可调用此前最近样本，但这些历史样本不进入输出。

以第 l 个样本结尾，取 $n_{i,l} = \min(W_{\mathrm{win}}, l)$ 个样本并划分为前、后半窗 $\mathcal{W}_{i,l}^{\mathrm{old}}$ 与 $\mathcal{W}_{i,l}^{\mathrm{new}}$，其大小分别为 $\lfloor n_{i,l}/2 \rfloor$ 和 $n_{i,l} - \lfloor n_{i,l}/2 \rfloor$；不足两个样本时保持上一状态。两半窗中心为

$$\bar{\boldsymbol{p}}_{i,l}^{q} = \frac{1}{|\mathcal{W}_{i,l}^{q}|} \sum_{h \in \mathcal{W}_{i,l}^{q}} \boldsymbol{p}_i^{t_h}, \qquad q \in \{\mathrm{old}, \mathrm{new}\}. \tag{A.2}$$

以前后半窗中心的欧氏距离衡量窗口内活动中心的平移量：

$$d_{i,l} = \left\| \bar{\boldsymbol{p}}_{i,l}^{\mathrm{new}} - \bar{\boldsymbol{p}}_{i,l}^{\mathrm{old}} \right\|_2. \tag{A.3}$$

$d_{i,l}$ 衡量活动中心迁移而非瞬时速度，因此围绕固定区域盘旋也可判为驻留。

为避免 $d_{i,l}$ 在单一阈值附近波动造成运动状态频繁切换，设置驻留进入阈值 δ_i^{in} 和驻留离开阈值 δ_i^{out}，并满足 $\delta_i^{\mathrm{in}} < \delta_i^{\mathrm{out}}$。令 $m_{i,l} \in \{\mathrm{stay}, \mathrm{move}\}$ 为第 l 个样本对应的运动状态，则滞回更新规则为

$$m_{i,l} = \begin{cases} \mathrm{move}, & m_{i,l-1} = \mathrm{stay} \ \wedge \ d_{i,l} > \delta_i^{\mathrm{out}}, \\ \mathrm{stay}, & m_{i,l-1} = \mathrm{move} \ \wedge \ d_{i,l} \le \delta_i^{\mathrm{in}}, \\ m_{i,l-1}, & \text{其他情况}. \end{cases} \tag{A.4}$$

最早样本初始化为 stay，后续按式 (A.4)递推；艇机采用不同阈值，以适应速度和活动半径差异。

分类后，将本周期内运动状态 $m_{i,l}$ 与任务标签 $\omega_i^{t_l}$ 均相同的相邻样本合并。设

第 s 段的连续索引集合为 $\mathcal{L}_{i,s}$，首尾索引为

$$l_{i,s}^{\text{from}} = \min \mathcal{L}_{i,s}, \qquad l_{i,s}^{\text{to}} = \max \mathcal{L}_{i,s}.$$

记起止时刻为 $t_{i,s}^{\text{from}}$、$t_{i,s}^{\text{to}}$，位置为 $\boldsymbol{p}_{i,s}^{\text{from}}$、$\boldsymbol{p}_{i,s}^{\text{to}}$，统一任务标签为 $\omega_{i,s}$。移动段保留这些边界量与样本数：

$$\mathcal{S}_{i,s}^{\text{move}} = \left(t_{i,s}^{\text{from}}, t_{i,s}^{\text{to}}, \boldsymbol{p}_{i,s}^{\text{from}}, \boldsymbol{p}_{i,s}^{\text{to}}, \omega_{i,s}, |\mathcal{L}_{i,s}|\right). \tag{A.5}$$

对于驻留段，以段内平均位置和活动半径代替全部位置点：

$$\begin{aligned} \bar{\boldsymbol{p}}_{i,s} &= \frac{1}{|\mathcal{L}_{i,s}|} \sum_{l \in \mathcal{L}_{i,s}} \boldsymbol{p}_i^{t_l}, \\ \rho_{i,s} &= \max_{l \in \mathcal{L}_{i,s}} \left\| \boldsymbol{p}_i^{t_l} - \bar{\boldsymbol{p}}_{i,s} \right\|_2, \\ \mathcal{S}_{i,s}^{\text{stay}} &= \left(t_{i,s}^{\text{from}}, t_{i,s}^{\text{to}}, \bar{\boldsymbol{p}}_{i,s}, \rho_{i,s}, \omega_{i,s}, |\mathcal{L}_{i,s}|\right). \end{aligned} \tag{A.6}$$

其中，$\rho_{i,s}$ 为驻留活动半径；任务标签变化时即使运动状态不变也切分轨迹段。图 A.1给出压缩示例。

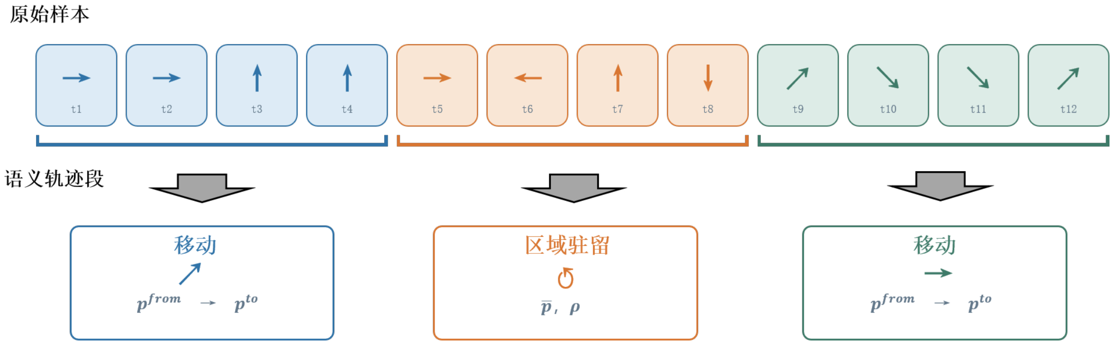

图 A.1: 平台原始轨迹的语义压缩示意图

系统同时提取区间 $(t_{k-1}^{\text{H}}, t_k^{\text{H}}]$ 内的发射、发现、毁伤和失效事件及其时刻、平台与目标关系。最终输出包含 t_k^{H} 的状态快照、语义轨迹段和增量事件，避免重复位置占用上下文。

附录 B　上层指挥决策器提示词模板

本附录列出五类大模型节点的提示词。$\langle\dot{\rangle}$ 由第 k 次上层决策的状态填充，集合、平台状态和事件均序列化为结构化文本；$\boldsymbol{\zeta}_{j,k}$ 表示第 j 个威胁中心。

B.1　威胁位置推断提示词

该模板用于已观测展开集群成员但核心位置未知的情况。

> **系统指令：**你是敌方集群核心位置推断专家。无人艇承担核心和主要火力载荷，无人机围绕其协同。仅依据已观测成员位置、类型、集群状态和编组知识推断核心；不得虚构平台或将初始情报当作当前位置。只输出结构化对象。
>
> **动态输入：**当前待推断威胁中心编号为 $\langle j\rangle$，集群状态为 ⟨集群状态⟩；已观测成员为 $\langle \mathcal{M}^{\mathrm{obs}}_{j,k}\rangle$，每个成员包含平台编号、平台类型和当前位置；已知敌方编组规律为 ⟨敌方艇机编组知识⟩。请估计集群核心位置。
>
> **输出格式：**{“position”:[x,y]}。

输出缺少有效二维坐标或调用失败时，程序以已观测成员的几何中心作为 $\widehat{\boldsymbol{p}}^{\mathrm{core}}_{j,k}$。

B.2　态势总结提示词

该模板从结构化上层状态中识别需调整任务分配的情况。

> **系统指令：**你是战场态势分析专家。你只能依据输入事实和给定知识进行判断，不得虚构距离、平台身份、因果关系或持续时间；证据不足时必须写“证据不足”，不得将可能性表述为已经发生的事实。
>
> 平台状态、压缩轨迹、事件、分组、威胁中心、当前数量和程序信号均为权威事实。威胁中心是推断记录而非环境真值，置信度达阈值仅表示可用于规划；紧急防守信号为真时必须增加阵位防守力量。
>
> 请逐项判断以下五种情况：
>
> 1. 未探明威胁数量大于当前执行覆盖搜索任务的无人机数量，是否需要增加侦察力量；
> 2. 是否出现新探明且尚未展开的威胁，并且需要增加主动拦截力量；
> 3. 敌方是否进入友方外沿且保护中心内侧防守力量不足；
> 4. 原主动拦截编组是否已完成攻击任务，需要转入防守或其他拦截任务；

> 5. 存活侦察平台是否完成了某一威胁的确认任务，可以减少覆盖搜索力量；
>
> 状态仅可为“成立”“不成立”或“证据不足”；成立时给出对应措施。依据须可追溯至输入，不得增设判断标准。只输出结构化对象。
>
> **动态输入：**战场知识为 ⟨战场背景与敌方编组知识⟩；任务知识为 ⟨三类任务定义⟩；情况判据为 ⟨任务调整判据⟩；结构化事实为 ⟨$\boldsymbol{s}_k^{\mathrm{H}}$⟩。
>
> **输出格式：** {“judgments”:[{“situation”: 编号，“status”:“成立 | 不成立 | 证据不足”，“judgment”:“情况判断”，“evidence”:[“直接事实依据”]}]，“solutions”:[{“situation”: 编号，“measure”:“对应措施”}]}。

核查未通过时追加上一版草案与核查问题，要求仅作最小修正，不引入新判断。

B.3 事实核查提示词

该节点仅核查事实，不重新决策。

> **系统指令：**你是有限权限的事实核查员，仅识别会改变成立状态或措施的硬错误，不进行战术规划。
>
> 仅报告两类问题：草案中的平台编号、数量、状态、事件或位置与原始事实矛盾；成立状态或措施违反显式判据。每项须给出事实位置、决策影响和最小修正。
>
> 不得新增占用关系、部署或拦截要求、阈值和评价维度，也不得因措辞、其他解释或保守偏好判定失败。若无影响决策的问题，判定通过且问题列表为空；只输出核查结果。
>
> **动态输入：**事实字段定义为 ⟨字段语义⟩；任务调整判据为 ⟨显式判据⟩；原始结构化事实为 ⟨$\boldsymbol{s}_k^{\mathrm{H}}$⟩；待核查的态势总结草案为 ⟨总结草案⟩。
>
> **输出格式：** {“passed”: 真或假，“issues”:[{“situation”: 编号，“type”:“事实矛盾或规则违反”，“fact_paths”:[“原始事实位置”]，“decision_impact”: 真或假，“problem”:“问题说明”，“required_revision”:“最小修正”}]}。

核查未通过时仅允许总结节点修正一次；输出无法解析则保留原草案并记录异常。

B.4 结论封装提示词

该节点将已核查内容压缩为资源规划所需信息，不新增判断。

> **系统指令：**你是态势结论编辑。保留资源规划所需的成立情况、关键数量、威胁中心编号、依据和措施，删除重复及无关推理；不得新增事实。证据不足时保守表述，不得把推测写成事实；仅保留成立情况对应的措施，无情况成立则保持当前分配。只输出结构化对象。

动态输入： 修正后的态势总结为 ⟨总结草案⟩；最新事实核查结果为 ⟨核查结果⟩。

输出格式： {“condition”:“精简后的情况”, “evidence”:“支持判断的关键事实”, “solution”:“对应措施”}。

B.5　资源规划提示词

该节点只确定三类任务的资源数量与拦截对象，不绑定平台编号。

系统指令： 你是无人集群资源规划专家。依据上一节点的情况、依据和措施，调整三类任务的平台数量并选择主动拦截中心。

必须遵守以下约束：三类任务的平台数量均为非负整数，且数量之和等于当前可用平台总数；覆盖搜索任务只能使用无人机，其数量不得超过可用无人机数量；主动拦截数量大于零时，必须存在可处置的高置信度威胁中心，并满足给定的无人艇—无人机编成比例；无可处置威胁时，主动拦截数量必须为零；紧急防守信号为真时，优先增加阵位防守数量；拦截对象只能从输入给出的候选威胁中心中选择。

结构化数量、候选中心和安全信号为权威事实，与自然语言结论冲突时以前者为准。不得指定平台编号；调整摘要用一句话说明变化方向。只输出结构化对象。

动态输入： 上一节点结论为 ⟨情况、依据与措施⟩；可用平台数量为 $\langle N_k^{\mathrm{U}}, N_k^{\mathrm{S}}\rangle$；当前任务数量为 ⟨三类任务当前数量⟩；威胁态势为 $\langle \mathcal{C}_k \rangle$；候选拦截对象为 ⟨达到置信度阈值且仍有效的威胁中心⟩；任务触发原因和安全约束为 ⟨触发信号与编成约束⟩。

输出格式： {“task_counts”:{“recon”: 整数, “attack”: 整数, “defend”: 整数}, “attack_targets”:[威胁中心编号], “adjustment”:“任务数量调整摘要”}。

程序校验数量守恒、平台资格、目标有效性和艇机比例。首次失败时追加原结果与违反约束，要求保持战术意图并仅修正违规项；再次失败则启用确定性规则。

B.6　任务绑定节点规则

任务绑定节点将资源数量和主动拦截对象确定性转换为平台级联合动作。设可用平台及无人机集合分别为 $\mathcal{N}_k$ 和 $\mathcal{N}_k^{\mathrm{U}}$，资源规划结果为

$$\boldsymbol{n}_k^{\mathrm{task}} = \left(n_k^{\mathrm{rec}}, n_k^{\mathrm{atk}}, n_k^{\mathrm{def}}\right), \qquad n_k^{\mathrm{rec}} + n_k^{\mathrm{atk}} + n_k^{\mathrm{def}} = |\mathcal{N}_k|, \tag{B.1}$$

以 $\mathcal{C}_k^{\mathrm{atk}} \subseteq \mathcal{C}_k$ 表示拦截对象。平台选择优先保持上一周期策略，再按到相关威胁中心的距离和编号排序。定义

$$\delta_{i,k}^q = \begin{cases} 0, & \omega_{i,k-1} = \omega^q, \\ 1, & \omega_{i,k-1} \neq \omega^q, \end{cases} \qquad q \in \{\mathrm{rec}, \mathrm{atk}\}, \tag{B.2}$$

$d_{i,k}^{\mathrm{rec}}$ 和 $d_{i,k}^{\mathrm{atk}}$ 分别为平台 i 到最近低置信度中心和指定拦截中心的距离。按 $(\delta_{i,k}^q, d_{i,k}^q, i)$ 升序排列，可依次保证策略连续性、距离和编号优先级。

主动拦截编组中无人机和无人艇的数量分别记为 $n_k^{\mathrm{A,U}}$ 和 $n_k^{\mathrm{A,S}}$。在资源允许时，编组满足

$$n_k^{\mathrm{A,U}} + n_k^{\mathrm{A,S}} = n_k^{\mathrm{atk}}, \qquad 2n_k^{\mathrm{A,S}} \leq n_k^{\mathrm{A,U}} \leq 3n_k^{\mathrm{A,S}}, \tag{B.3}$$

即每艘火力核心无人艇配置 2 至 3 架无人机，优先接近 2:1；资源不足以合法编组时直接采用安全兜底。完整过程见算法 1。

该顺序保证覆盖搜索资格和主动拦截编成约束；三个集合须两两不交、并集为 $\mathcal{N}_k$ 且规模符合规划，否则执行全员阵位防守。

附录 C　覆盖搜索热点方向补偿方法

当无人机进入已充分搜索或低概率区域时，八个方向的新增熵减均可能接近零，局部信息增益无法提供脱离方向。热点补偿仅在此类低信息区域提供辅助引导。

对于无人机 i 的候选动作 a，以动作后探测范围相对于当前位置探测范围新增的网格集合为 $\Delta\mathcal{B}_i^t(a)$，对应的潜在新增信息量为

$$I_i^t(a) = \sum_{(x_{\mathrm{g}}, y_{\mathrm{g}}) \in \Delta\mathcal{B}_i^t(a)} H_{x_{\mathrm{g}}, y_{\mathrm{g}}}^t. \tag{C.1}$$

若八个方向的 $I_i^t(a)$ 均低于阈值，则提取概率图中的高概率网格，经连通域合并并滤除小面积区域。有效热点 $\mathcal{V}_m^{\mathrm{hot}}$ 的概率加权中心为

$$\boldsymbol{c}_m^t = \frac{\displaystyle\sum_{(x_{\mathrm{g}}, y_{\mathrm{g}}) \in \mathcal{V}_m^{\mathrm{hot}}} p_{x_{\mathrm{g}}, y_{\mathrm{g}}}^t \boldsymbol{c}_{x_{\mathrm{g}}, y_{\mathrm{g}}}}{\displaystyle\sum_{(x_{\mathrm{g}}, y_{\mathrm{g}}) \in \mathcal{V}_m^{\mathrm{hot}}} p_{x_{\mathrm{g}}, y_{\mathrm{g}}}^t}, \tag{C.2}$$

选择最近的热点中心作为补偿目标 $\boldsymbol{c}_i^{t,*}$，其几何关系见图 C.1。

信息阈值、高概率分割阈值和最小连通域面积分别用于判定低信息状态、提取热点和滤除噪声。

算法 1 上层任务绑定与安全校验

```
Require: 可用平台 $\mathcal{N}_k$ 及其类型、位置和上一周期任务；任务数量 $\boldsymbol{n}_k^{\text{task}}$；主动拦截对象 $\mathcal{C}_k^{\text{atk}}$；威胁中心集合 $\mathcal{C}_k$
Ensure: 平台级联合任务 $\boldsymbol{\omega}_k$ 及三类任务平台集合 $\mathcal{I}_{\text{rec},k}$、$\mathcal{I}_{\text{atk},k}$ 和 $\mathcal{I}_{\text{def},k}$
初始化三个任务平台集合为空集
if 任务数量不满足式 (B.1)，或 $n_k^{\text{rec}} > |\mathcal{N}_k^{\text{U}}|$ then
    将全部平台分配至 $\mathcal{I}_{\text{def},k}$，并返回安全防守任务
end if
if $n_k^{\text{atk}} > 0$ 且 $\mathcal{C}_k^{\text{atk}}$ 为空，或集合中存在无效威胁中心 then
    将全部平台分配至 $\mathcal{I}_{\text{def},k}$，并返回安全防守任务
end if
按 $(\delta_{i,k}^{\text{rec}}, d_{i,k}^{\text{rec}}, i)$ 升序排列 $\mathcal{N}_k^{\text{U}}$
取前 $n_k^{\text{rec}}$ 个平台构成 $\mathcal{I}_{\text{rec},k}$
$\mathcal{N}_k^{\text{rem}} \leftarrow \mathcal{N}_k \setminus \mathcal{I}_{\text{rec},k}$
枚举满足式 (B.3)且不超过 $\mathcal{N}_k^{\text{rem}}$ 艇机资源的可行编成集合 $\mathcal{F}_k$
if $n_k^{\text{atk}} > 0$ 且 $\mathcal{F}_k = \varnothing$ then
    将全部平台分配至 $\mathcal{I}_{\text{def},k}$，并返回安全防守任务
end if
if $n_k^{\text{atk}} > 0$ then
    从 $\mathcal{F}_k$ 中选取最接近 2:1 且保留较多无人艇火力核心的编成 $(n_k^{\text{A,U}}, n_k^{\text{A,S}})$
    分别按 $(\delta_{i,k}^{\text{atk}}, d_{i,k}^{\text{atk}}, i)$ 升序排列剩余无人机与无人艇
    选择前 $n_k^{\text{A,U}}$ 架无人机和前 $n_k^{\text{A,S}}$ 艘无人艇构成 $\mathcal{I}_{\text{atk},k}$
end if
$\mathcal{I}_{\text{def},k} \leftarrow \mathcal{N}_k \setminus (\mathcal{I}_{\text{rec},k} \cup \mathcal{I}_{\text{atk},k})$
根据三个平台集合生成 $\boldsymbol{\omega}_k$，并附加主动拦截对象 $\mathcal{C}_k^{\text{atk}}$
检查平台可用性、分配唯一性、分配完备性、任务数量、平台资格、目标有效性和艇机编成
if 任一检查未通过 then
    $\mathcal{I}_{\text{rec},k}, \mathcal{I}_{\text{atk},k} \leftarrow \varnothing$，$\mathcal{I}_{\text{def},k} \leftarrow \mathcal{N}_k$
    生成全员阵位防守的安全任务 $\boldsymbol{\omega}_k$
end if
return $\boldsymbol{\omega}_k$，$\mathcal{I}_{\text{rec},k}$，$\mathcal{I}_{\text{atk},k}$，$\mathcal{I}_{\text{def},k}$
```

令无人机依据动作 a_i^t 形成的移动向量为

$$\boldsymbol{v}_{i,\text{a}}^t = \boldsymbol{p}_i^{t,\text{next}} - \boldsymbol{p}_i^t, \tag{C.3}$$

由当前位置指向热点中心的参考向量为

$$\boldsymbol{v}_{i,\text{h}}^t = \boldsymbol{c}_i^{t,*} - \boldsymbol{p}_i^t. \tag{C.4}$$

两者夹角为

$$\theta_{i,\text{hot}}^t = \arccos\left(\frac{(\boldsymbol{v}_{i,\text{a}}^t)^{\text{T}} \boldsymbol{v}_{i,\text{h}}^t}{\|\boldsymbol{v}_{i,\text{a}}^t\|_2 \|\boldsymbol{v}_{i,\text{h}}^t\|_2}\right). \tag{C.5}$$

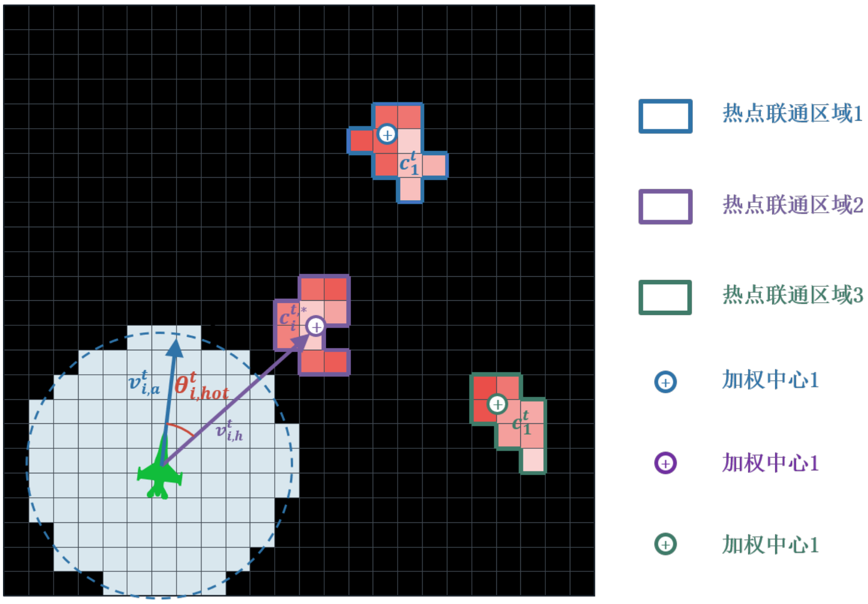
c_1^t
$c_i^{t,*}$
$v_{i,a}^t$
$\theta_{i,hot}^t$
$v_{i,h}^t$
c_1^t
热点联通区域1
热点联通区域2
热点联通区域3
加权中心1
加权中心1
加权中心1

图 C.1: 热点连通域、概率加权中心与方向夹角示意

热点方向补偿奖励定义为

$$r_{i,\mathrm{hot}}^t = \lambda_{\mathrm{hot}}\,\mathrm{clip}\left(\frac{\pi}{2} - \theta_{i,\mathrm{hot}}^t, -\frac{\pi}{2}, \frac{\pi}{2}\right), \tag{C.6}$$

其中 $\lambda_{\mathrm{hot}} > 0$。朝向、垂直和背离热点分别获得正、零和负奖励；低信息状态持续时保持目标以抑制振荡，恢复有效信息增益后停止补偿，不改变熵减奖励的主体地位。

附录 D　原始数据附表

表 D.1–表 D.6记录第三至第五章主要多场景对比实验的逐回合数据。Cfg 为场景或参数配置，M 为方法，K 为指标；第三、第四章表头的 0–49 为同一配置下的独立测试序号，第五章表头的 31–60 为配对随机种子。同一场景、方法与指标的一行数据对应正文箱线图和统计检验使用的完整回合样本，未事后删除极端值。

指标口径与正文保持一致。第三章的目标发现比例取值为 $[0, 1]$，目标发现时间以决策步计，未发现目标按单回合最大决策步计入，回合平均信息熵取回合内全局信息熵在各决策步及其仿真子步上的平均值。第四、第五章的胜率、拦截率与损失率在附表中以 $[0, 1]$ 的比例记录；弹药利用效率为拦截数与弹药消耗量之比，平均拦截距离为拦截位置到保护中心的平均欧氏距离。回合奖励为对应章节所定义团队奖励的逐步累加，仅用于同一实验设置内的方法对比。各表的方法缩写和指标字母另在表后注释。

表 D.1: 第三章搜索算法多场景测试原始数据

Cfg	M	K	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49
t12r3	D	N	0.750	0.583	0.667	0.667	0.583	0.833	0.583	0.833	0.583	0.833	0.667	0.667	0.667	0.833	0.667	0.667	0.500	0.583	0.500	0.750	0.583	0.583	0.917	0.583	0.833	0.833	0.667	0.750	0.417	0.750	0.917	0.583	0.833	0.750	0.667	0.833	0.750	0.750	0.583	0.667	0.583	0.500	0.750	0.667	0.583	0.500	0.750	0.667	0.917	0.583
	D	T	34.833	39.417	36.000	38.917	40.500	40.750	40.750	33.333	41.167	25.750	39.833	36.000	38.500	32.667	40.333	39.083	46.500	42.417	47.833	39.167	44.583	43.250	23.500	47.417	26.833	34.833	39.000	33.500	51.417	34.667	23.333	44.917	28.167	34.583	41.083	30.417	29.000	36.917	39.000	39.750	38.583	49.750	35.917	41.083	45.000	48.500	40.167	35.250	34.250	45.667
	D	H	192.836	199.967	213.256	263.238	153.863	252.227	202.156	225.057	194.386	178.691	224.894	203.409	241.058	221.906	222.223	246.492	248.375	229.303	220.802	192.981	191.385	205.383	214.483	233.632	193.688	233.088	245.770	247.425	225.985	176.949	183.858	232.591	210.056	219.476	212.344	190.705	184.747	228.956	262.808	229.085	189.198	219.332	189.134	217.740	215.459	196.786	172.790	193.314	242.829	244.917
	M	N	0.667	0.417	0.583	0.750	0.417	0.583	0.500	0.583	0.833	0.417	0.750	0.583	0.500	0.583	0.667	0.750	0.667	0.500	0.500	0.583	0.333	0.667	0.417	0.833	0.583	0.250	0.500	0.750	0.583	0.750	0.417	0.500	0.833	0.750	0.667	0.667	0.500	0.667	0.833	0.417	0.750	0.417	0.750	0.667	0.667	0.667	0.583	1.000	0.667	0.583
	M	T	44.583	61.417	43.167	40.250	56.083	51.250	68.417	59.417	41.000	58.083	46.667	49.917	58.083	47.917	40.333	47.000	47.333	58.000	51.083	44.667	69.000	55.333	54.417	43.167	41.833	64.750	56.250	45.167	50.833	51.333	56.333	54.583	45.833	41.250	48.167	42.833	52.833	42.917	34.583	56.917	34.750	63.333	50.750	48.833	41.500	55.000	46.500	46.750	48.917	52.500
	M	H	245.205	265.338	257.719	244.399	218.521	258.381	275.158	237.377	234.946	259.012	209.611	245.941	265.483	246.365	244.602	250.235	264.845	285.841	221.148	235.951	269.983	238.225	235.070	269.807	236.553	211.267	268.129	288.699	252.091	248.350	236.397	240.909	279.160	216.804	253.130	263.067	253.875	260.402	234.595	248.642	248.662	292.331	250.704	307.882	241.496	265.098	282.956	257.574	224.834	227.072
	B	N	0.667	0.500	0.417	0.667	0.667	0.500	0.750	0.333	0.417	0.417	0.417	0.667	0.250	0.083	0.500	0.500	0.750	0.500	0.333	0.333	0.167	0.250	0.500	0.583	0.250	0.667	0.417	0.583	0.583	0.417	0.667	0.667	0.417	0.500	0.417	0.333	0.583	0.333	0.583	0.333	0.417	0.583	0.833	0.583	0.500	0.500	0.583	0.417	0.333	0.750
	B	T	50.917	62.667	56.167	52.833	43.667	64.667	36.000	61.750	58.917	61.333	67.500	64.333	63.917	75.333	63.083	65.500	40.167	47.750	69.083	66.333	68.083	65.333	48.583	45.917	68.667	52.500	56.500	54.000	55.583	64.833	49.083	50.583	66.917	57.000	57.500	58.167	55.833	58.417	56.417	62.333	66.750	58.083	47.417	53.667	61.583	55.917	53.500	60.167	61.083	49.833
	B	H	237.766	296.468	284.530	242.346	268.848	317.587	237.088	288.222	315.463	299.685	314.771	304.354	323.490	324.549	320.662	322.635	238.873	266.999	322.652	319.112	323.661	310.840	306.352	234.389	336.886	233.208	267.741	268.958	276.776	256.589	271.546	271.572	330.566	266.191	287.853	303.129	237.418	330.768	235.618	315.902	323.929	270.494	231.392	292.141	310.391	284.413	306.111	283.684	329.051	236.158
	I	N	0.833	0.833	0.750	0.583	0.833	0.667	0.667	0.833	0.667	0.750	0.583	0.667	0.667	0.917	0.750	0.833	0.833	0.583	0.583	0.917	0.750	0.833	1.000	0.667	0.833	0.750	0.667	0.750	0.750	0.917	0.917	0.833	0.583	0.833	0.750	0.833	0.833	0.750	0.500	0.667	0.500	0.917	0.833	0.667	0.750	0.833	0.667	0.750	0.833	0.833
	I	T	38.333	48.250	44.583	52.000	35.000	39.000	42.500	35.667	43.750	51.750	44.000	50.833	44.167	33.000	40.917	30.750	39.667	42.250	65.167	31.000	48.167	45.583	25.417	40.083	42.250	28.167	41.500	39.667	52.500	31.500	31.167	32.000	46.333	33.667	47.167	32.167	38.667	37.333	49.500	50.500	54.833	32.500	27.333	43.750	53.833	42.583	43.500	43.167	26.250	41.250
	I	H	195.790	231.838	210.467	215.304	189.593	266.891	212.707	229.841	228.472	230.870	256.399	245.744	227.435	216.574	241.038	210.802	165.443	192.865	252.880	195.237	209.883	216.879	213.670	250.242	177.590	243.139	164.580	235.047	167.406	220.735	201.887	209.170	185.375	199.058	235.443	206.038	248.230	228.140	250.547	235.221	232.549	227.070	224.763	179.486	235.791	180.072	270.200	223.727	230.778	199.594
	P	N	0.917	0.750	0.500	0.667	0.917	0.833	0.750	0.917	0.667	0.500	0.750	0.667	0.500	0.750	0.667	0.833	0.750	0.917	0.750	0.750	0.833	0.750	0.833	0.583	0.500	0.917	0.750	0.917	0.833	0.750	0.500	0.750	0.750	0.667	0.500	0.667	0.750	0.917	0.833	0.667	0.750	0.667	0.833	0.667	0.750	0.667	0.583	0.833	0.667	0.667
	P	T	30.000	53.833	51.417	41.333	34.250	36.750	43.250	33.000	52.083	51.167	42.000	39.833	49.417	41.083	41.917	42.750	39.250	41.583	39.500	31.500	41.750	41.083	33.000	51.750	58.000	27.167	45.917	24.333	29.000	44.667	48.750	47.750	38.917	43.500	57.250	51.000	37.583	34.083	38.583	49.333	41.000	50.667	49.333	46.000	35.250	52.750	46.083	36.417	37.583	49.250
	P	H	211.116	190.276	235.292	249.727	236.822	220.021	205.071	169.419	227.612	187.825	232.281	253.389	243.851	204.731	181.592	216.527	213.352	237.409	184.545	230.890	225.494	195.139	178.114	203.051	293.903	239.368	220.755	201.484	205.832	245.981	281.000	216.490	278.444	231.838	232.302	203.219	218.922	205.202	228.859	228.984	243.233	201.898	216.212	224.227	190.240	254.900	252.991	193.321	218.364	233.617
	T	N	0.417	0.833	0.833	0.333	0.417	0.750	0.750	0.750	0.667	0.750	0.750	0.750	0.833	0.833	0.500	0.833	0.583	0.750	0.667	0.750	0.667	0.583	0.833	0.583	0.500	0.833	0.500	0.750	0.583	0.833	0.833	0.667	0.667	0.667	0.917	0.750	0.667	0.750	0.750	0.833	0.583	0.750	0.500	0.667	0.667	0.750	0.750	0.500	0.667	0.833
	T	T	53.333	43.417	48.333	56.333	58.000	41.583	40.333	46.917	39.167	30.250	36.667	42.333	45.167	28.167	53.750	36.083	42.750	39.500	39.083	41.667	46.250	50.833	35.667	50.667	43.750	36.083	55.667	35.917	51.833	32.500	31.083	40.167	51.167	52.000	34.167	46.250	51.167	44.333	38.833	37.417	44.417	38.250	52.000	47.750	43.500	43.083	41.500	58.333	44.000	41.417
	T	H	263.975	240.877	212.425	193.645	260.953	256.101	183.739	257.818	182.802	231.080	218.863	240.341	200.791	182.383	249.937	202.060	202.871	207.121	214.420	207.078	215.728	230.492	207.432	209.412	217.349	186.155	245.856	190.687	215.530	200.540	258.247	232.411	211.173	225.058	181.453	221.692	258.294	236.831	209.833	235.181	224.397	233.246	225.537	184.747	252.231	259.376	216.439	234.823	208.312	213.973
t12r6	D	N	0.750	0.833	0.917	0.833	0.750	0.917	0.917	0.667	0.833	1.000	0.750	0.917	0.833	0.833	1.000	0.667	0.917	0.917	0.917	0.917	0.667	0.833	0.833	0.750	1.000	0.750	1.000	0.750	0.833	0.833	0.833	0.833	0.833	1.000	1.000	0.917	1.000	0.917	1.000	1.000	0.917	0.833	0.750	0.750	0.833	0.667	0.833	1.000	0.917	0.917
	D	T	28.750	35.000	33.833	30.750	35.000	30.000	25.083	34.083	30.167	16.667	33.083	24.750	32.083	27.583	26.917	36.917	24.917	19.833	25.083	29.083	39.500	26.667	29.917	36.917	16.250	35.250	22.750	31.917	27.083	34.250	31.667	25.333	22.250	23.083	20.250	23.417	24.000	22.833	22.083	11.583	22.083	27.083	32.000	30.167	32.667	35.333	26.833	22.250	21.750	22.667
	D	H	186.235	216.280	224.171	174.725	177.151	167.591	124.657	203.942	197.071	214.070	176.938	149.883	190.485	164.871	170.936	185.587	161.771	186.102	167.054	180.219	215.924	185.757	205.905	156.693	129.101	167.079	205.436	198.236	184.090	181.158	154.677	212.341	190.396	191.923	184.021	144.038	220.661	183.296	196.139	187.711	188.017	195.021	163.088	170.918	189.579	163.805	165.042	197.249	173.055	185.722
	M	N	0.500	0.583	0.750	0.750	0.750	0.833	0.500	0.583	0.667	0.917	0.583	0.833	0.833	0.917	0.583	0.500	0.917	0.750	0.667	0.833	0.250	0.917	0.417	0.750	0.583	0.667	0.917	0.667	0.917	0.750	0.750	0.833	0.583	0.417	0.917	0.833	1.000	0.667	0.750	0.583	0.833	0.750	0.667	0.750	0.583	0.833	1.000	0.417	0.750	0.667
	M	T	63.500	52.750	53.583	39.417	38.167	30.667	50.417	53.750	49.167	34.000	60.417	47.250	37.833	43.833	50.500	55.250	41.417	50.750	49.333	43.250	65.083	44.000	65.667	42.583	50.833	58.250	54.583	47.833	30.750	48.417	46.333	45.250	47.833	60.250	31.500	39.917	29.583	51.833	40.917	45.500	40.750	49.250	50.917	47.750	43.583	37.000	28.583	57.833	52.250	52.750
	M	H	241.292	230.075	226.682	197.846	221.691	214.388	250.863	226.364	225.960	252.880	222.102	243.630	238.644	218.850	238.014	276.488	221.211	190.546	257.711	170.158	258.982	190.777	223.348	235.651	253.580	228.851	263.605	221.764	186.576	250.456	232.152	241.536	246.695	249.653	210.061	227.410	222.776	277.462	219.237	237.728	208.359	240.986	220.173	220.898	227.192	242.063	183.096	202.025	267.907	270.274
	B	N	1.000	0.417	0.750	0.833	0.417	0.500	0.833	0.833	0.833	0.417	0.667	0.417	0.750	0.667	0.500	0.833	0.750	0.750	0.917	0.833	0.750	1.000	0.583	0.500	0.667	1.000	0.333	0.167	0.333	0.750	0.917	0.917	0.917	0.583	0.167	0.167	0.333	0.833	0.583	0.917	0.417	1.000	0.500	0.417	0.583	0.917	0.833	0.667	0.750	0.750
	B	T	29.167	53.083	50.833	42.917	63.000	49.083	33.917	40.500	40.417	54.083	40.333	54.500	45.500	45.917	55.167	56.417	40.917	59.000	31.167	44.583	40.500	31.250	53.083	55.083	43.417	30.833	63.583	72.250	61.500	49.833	46.167	31.917	40.917	48.250	68.667	74.083	69.917	52.667	50.500	31.917	61.667	27.917	49.250	59.750	51.750	28.667	35.667	55.667	44.500	47.500
	B	H	180.360	249.641	249.385	211.238	239.535	190.560	187.206	219.429	208.134	210.285	221.050	191.025	246.582	261.889	289.595	290.936	239.161	273.134	202.456	218.001	237.577	166.749	234.720	187.149	216.070	201.739	265.748	316.867	284.915	256.296	233.979	240.704	214.532	183.759	329.865	307.847	308.983	245.792	234.474	244.672	280.487	218.598	177.873	275.471	218.688	184.789	200.183	233.078	214.280	223.016
	I	N	0.833	0.750	0.917	0.833	1.000	1.000	0.833	1.000	1.000	1.000	0.917	1.000	0.833	1.000	0.750	0.917	0.833	0.833	0.917	0.833	0.833	0.833	0.833	0.917	0.833	0.917	0.833	0.833	1.000	1.000	1.000	0.750	0.917	1.000	1.000	0.917	0.917	0.917	0.833	0.833	0.833	0.833	0.917	0.833	0.833	0.917	1.000	1.000	0.833	0.667
	I	T	27.917	37.583	27.917	30.917	29.583	13.333	31.917	22.083	23.750	18.417	30.417	26.250	33.083	19.333	33.333	34.000	33.667	32.583	22.750	30.667	33.500	26.333	38.250	29.917	28.833	20.417	23.667	34.167	17.167	18.583	23.333	34.250	33.417	20.667	17.833	30.417	24.917	27.167	28.667	30.417	29.917	42.917	22.250	27.167	27.750	20.417	23.167	21.333	27.000	41.250
	I	H	162.289	162.818	190.801	190.089	183.944	209.960	157.710	162.638	129.224	134.920	152.827	185.942	143.539	169.627	136.956	160.997	191.178	155.089	171.217	159.888	180.804	175.763	166.993	155.114	141.581	125.216	159.359	216.875	183.239	194.477	171.476	140.781	143.786	177.466	198.668	140.298	148.719	154.083	132.282	187.552	141.419	158.803	183.754	181.013	114.566	161.102	167.401	166.974	144.453	193.799
	P	N	0.833	0.750	0.833	0.750	0.833	0.833	0.833	0.917	0.833	1.000	0.833	1.000	0.917	0.917	1.000	0.750	1.000	0.833	0.917	1.000	0.750	0.833	0.833	0.833	0.917	0.833	0.833	0.750	1.000	1.000	0.833	0.917	0.833	0.917	0.833	0.833	0.750	0.750	0.667	0.833	1.000	0.667	0.917	0.917	0.750	1.000	0.833	1.000	0.667	0.833
	P	T	36.667	37.333	44.667	49.167	43.667	27.667	36.917	28.583	28.667	24.083	31.000	26.083	31.083	35.083	31.417	44.167	24.750	34.250	31.583	33.083	30.167	45.250	39.000	33.833	33.417	33.917	33.500	39.500	27.167	29.583	37.000	28.167	37.583	27.833	32.000	41.083	38.417	45.500	53.500	35.417	26.750	48.250	33.083	25.417	44.917	31.583	32.833	31.250	44.917	36.417
	P	H	197.411	217.496	212.375	246.674	199.710	191.838	147.603	178.820	172.897	190.669	177.130	201.150	140.623	194.319	193.964	182.963	159.483	192.453	174.036	170.126	148.947	195.591	174.101	165.705	157.690	175.267	170.278	196.588	136.955	171.248	176.672	188.213	221.511	164.452	145.616	176.098	222.106	178.988	171.609	196.461	146.760	220.087	178.092	187.307	220.618	185.866	150.973	173.005	214.636	225.994
	T	N	0.917	0.750	0.917	0.917	0.667	0.750	0.833	0.833	0.917	0.917	0.750	0.917	0.750	0.917	0.750	0.833	1.000	0.917	1.000	0.917	0.917	1.000	0.750	0.917	0.917	0.833	0.833	0.750	0.917	0.833	0.833	0.750	0.917	0.917	0.750	1.000	0.833	0.750	0.750	0.917	0.833	0.833	0.833	0.667	0.750	0.750	0.667	0.750	0.833	1.000
	T	T	38.917	41.417	31.500	28.667	52.500	43.667	39.833	37.417	22.417	28.167	32.500	29.250	37.417	24.667	31.583	34.000	32.167	28.500	28.583	46.583	34.000	20.000	35.250	35.000	37.833	36.083	33.667	36.167	29.250	33.667	41.417	38.750	34.917	31.083	38.833	34.250	39.417	40.167	38.667	34.000	44.917	36.833	40.500	39.500	36.000	53.000	43.500	44.917	44.083	23.000
	T	H	227.723	181.964	164.793	179.251	182.990	177.587	165.268	149.097	162.028	195.549	130.998	228.617	179.562	205.446	196.185	230.161	198.364	194.574	202.238	172.748	211.602	200.812	195.835	149.968	211.315	184.542	162.943	190.648	186.073	172.980	240.253	173.524	202.517	212.840	229.118	191.264	203.068	220.036	170.430	175.513	172.115	205.013	186.913	213.322	170.429	193.878	210.358	150.530	180.230	213.552
t15r3	D	N	0.867	0.800	0.600	0.533	0.733	0.800	0.800	0.667	0.600	0.200	0.667	0.800	0.800	0.600	0.667	0.533	0.600	0.667	0.600	0.733	0.400	0.667	0.600	0.867	0.667	0.733	0.733	0.667	0.600	0.667	0.600	0.667	0.600	0.867	0.800	0.867	0.800	0.600	0.733	0.600	0.533	0.733	0.600	0.467	0.667	0.533	0.600	0.600	0.667	0.467
	D	T	25.667	30.067	44.400	43.667	33.933	28.333	29.067	42.467	41.800	65.000	39.267	31.133	27.667	43.533	38.667	44.867	38.133	45.600	40.667	36.867	51.733	41.533	43.000	25.133	39.333	35.133	33.467	38.200	41.733	42.333	52.533	40.533	47.267	27.667	31.533	25.867	32.800	42.933	30.733	41.533	52.133	33.333	42.933	50.467	35.267	45.333	46.533	42.667	41.933	48.333
	D	H	237.244	236.728	269.184	301.616	253.111	222.775	290.094	270.574	272.819	329.877	250.956	272.107	219.442	209.525	269.289	273.427	338.879	244.169	257.513	307.861	325.024	275.841	254.772	207.621	270.440	255.764	296.433	235.691	281.874	280.137	269.090	226.059	271.754	242.433	235.673	220.034	243.659	251.563	290.722	303.974	323.495	280.217	276.870	263.657	302.881	256.659	268.354	278.762	268.272	303.081
	M	N	0.467	0.533	0.667	0.467	0.400	0.533	0.533	0.733	0.533	0.733	0.600	0.933	0.733	0.600	0.600	0.600	0.600	0.533	0.733	0.533	0.467	0.600	0.533	0.333	0.400	0.533	0.533	0.533	0.600	0.667	0.600	0.667	0.733	0.800	0.600	0.400	0.467	0.400	0.600	0.467	0.533	0.533	0.800	0.533	0.867	0.667	0.800	0.400	0.733	0.467
	M	T	60.733	52.800	47.000	56.200	55.533	59.533	49.467	36.867	50.867	42.000	42.800	37.467	37.067	43.733	49.733	50.200	44.733	55.400	43.067	56.533	59.000	42.067	51.000	60.467	60.533	49.467	62.067	52.667	53.800	42.667	51.933	52.467	41.933	36.733	43.533	60.867	59.133	58.133	48.067	51.800	57.467	48.867	41.867	48.533	39.200	48.667	46.667	56.200	37.067	49.800
	M	H	287.898	257.616	286.784	273.939	321.829	253.771	275.452	241.375	260.680	260.373	297.196	254.961	262.929	297.154	331.786	243.869	254.895	304.016	265.625	282.279	300.473	240.084	322.405	323.395	241.957	333.786	303.932	280.829	329.045	285.091	343.239	325.663	271.898	252.223	263.706	275.113	320.266	255.185	285.099	275.861	286.598	263.057	312.801	253.190	285.793	296.179	291.076	293.742	291.913	293.010
	B	N	0.733	0.333	0.200	0.467	0.200	0.333	0.267	0.267	0.333	0.400	0.267	0.400	0.333	0.333	0.333	0.267	0.400	0.200	0.267	0.133	0.400	0.333	0.467	0.733	0.267	0.333	0.400	0.600	0.267	0.600	0.333	0.400	0.267	0.467	0.400	0.400	0.267	0.267	0.333	0.533	0.200	0.600	0.200	0.467	0.467	0.333	0.667	0.467	0.267	0.333
	B	T	59.200	64.933	71.600	52.733	68.667	61.267	67.133	63.533	58.800	63.400	68.600	57.200	57.267	62.267	64.867	66.333	59.733	67.200	62.000	69.800	56.133	59.600	60.467	45.533	63.933	61.733	56.667	45.067	67.400	51.533	63.467	61.067	64.067	58.733	57.533	59.800	70.000	62.600	63.600	54.267	67.400	53.200	67.733	65.133	56.600	60.333	54.800	65.067	69.400	64.400
	B	H	308.042	380.339	328.688	358.457	383.536	364.618	396.841	334.116	301.988	392.385	331.050	308.038	385.086	324.105	342.731	375.342	367.206	345.872	306.834	366.504	335.665	295.253	331.195	286.553	347.369	327.499	299.042	294.296	346.108	302.686	369.289	357.652	329.640	297.559	321.662	317.901	335.952	317.076	309.207	345.826	358.220	294.892	367.990	337.620	315.366	287.510	336.280	369.815	353.748	332.883
	I	N	0.733	0.467	0.467	0.600	0.533	0.733	0.533	0.733	0.800	0.600	0.667	0.800	0.733	0.733	0.667	0.733	0.733	0.467	0.533	0.733	0.533	0.800	0.867	0.533	0.800	0.600	0.733	0.733	0.867	0.600	0.933	0.400	0.933	0.800	0.533	0.800	0.733	0.733	0.867	0.600	0.867	0.533	0.733	0.733	0.533	0.733	0.533	0.733	0.800	0.600
	I	T	41.400	54.467	50.733	45.267	45.600	39.267	53.467	42.267	34.333	42.200	48.867	46.133	38.867	36.400	41.533	38.400	45.000	52.867	49.533	52.267	48.667	44.933	36.800	45.067	40.200	47.400	38.600	40.267	37.467	53.133	33.800	56.933	38.133	37.200	55.800	39.533	39.733	43.467	41.400	50.867	34.200	47.600	48.200	41.400	48.867	43.867	50.133	45.933	39.267	43.200
	I	H	294.656	268.648	279.199	279.870	267.161	274.224	237.719	284.631	242.457	251.452	275.221	278.854	261.682	227.901	260.314	237.728	263.801	285.280	293.876	269.065	275.896	274.752	223.423	236.013	252.947	294.777	242.286	273.433	214.732	274.949	299.021	323.640	205.050	258.272	253.866	269.010	250.323	198.242	251.435	284.375	281.281	294.961	237.888	221.484	256.129	190.392	264.696	213.032	264.515	241.572
	P	N	0.933	0.667	0.600	0.667	0.800	0.667	0.600	0.667	0.733	0.800	0.667	0.667	0.600	0.867	0.867	0.733	0.400	0.600	0.533	0.600	0.600	0.600	0.600	0.533	0.667	0.800	0.533	0.533	0.533	0.733	0.533	0.533	0.667	0.800	0.800	0.733	0.467	0.733	0.600	0.800	0.600	0.600	0.467	0.667	0.800	0.867	0.733	0.867	0.733	0.667
	P	T	32.067	41.133	49.067	40.600	46.800	39.533	45.267	49.400	40.333	34.267	38.933	35.733	49.533	30.733	33.600	40.800	58.267	47.467	51.133	53.000	45.667	40.200	43.867	42.667	45.400	40.800	49.733	48.533	49.800	38.733	51.333	51.200	50.867	41.600	37.800	44.933	52.067	40.200	41.067	41.067	38.733	45.533	47.600	41.800	35.200	33.733	35.533	33.733	34.800	42.933
	P	H	237.435	253.839	221.531	268.872	270.477	247.070	274.047	262.588	234.706	266.080	293.745	222.825	287.268	268.710	269.995	243.872	288.363	259.224	227.495	295.203	263.056	280.896	326.087	295.137	246.218	220.693	220.257	221.898	237.793	277.022	277.049	312.039	264.623	313.325	283.794	249.154	241.658	259.799	278.771	248.977	291.201	308.664	250.801	255.743	253.317	262.082	273.727	228.186	271.022	291.810
	T	N	0.600	0.733	0.533	0.600	0.533	0.467	0.600	0.600	0.733	0.733	0.733	0.733	0.867	0.667	0.533	0.600	0.733	0.733	0.667	0.533	0.467	0.600	0.533	0.800	0.800	0.667	0.600	0.667	0.600	0.467	0.733	0.667	0.533	0.667	0.667	0.867	0.600	0.600	0.467	0.400	0.667	0.600	0.667	0.467	0.600	0.667	0.533	0.600	0.667	0.600
	T	T	40.933	42.667	47.333	40.933	54.667	55.000	41.333	52.933	43.733	46.800	44.000	39.133	51.867	44.267	52.067	47.267	49.333	36.533	55.867	53.933	59.200	46.000	49.800	44.933	39.533	44.000	39.467	44.267	45.000	57.733	47.867	48.200	47.667	49.467	51.400	31.733	56.267	44.333	51.533	64.067	50.267	44.467	45.200	58.000	48.933	36.933	51.333	50.933	52.067	44.067
	T	H	283.702	265.816	314.823	255.418	318.957	277.284	268.684	306.436	283.877	303.572	220.377	251.738	292.091	292.396	232.948	248.515	277.081	288.375	292.163	248.510	249.807	250.444	252.453	291.652	263.723	266.019	230.100	261.402	225.651	294.725	231.771	308.713	268.728	290.599	298.228	200.607	273.681	308.785	299.776	318.109	295.401	261.616	261.015	296.739	317.432	258.849	262.313	262.150	261.539	265.751
t18r6	D	N	0.722	0.778	0.833	0.778	0.556	0.889	0.944	0.667	0.722	0.722	0.889	0.833	0.556	0.778	0.944	0.722	0.722	0.667	0.667	0.833	0.722	0.833	0.611	0.833	0.778	0.778	0.833	0.833	0.722	0.667	0.833	0.722	0.833	0.667	0.722	0.889	0.778	0.778	0.944	0.611	0.778	0.556	0.833	0.722	0.778	0.833	0.722	0.889	0.556	0.833
	D	T	35.667	30.500	27.833	36.611	45.833	19.333	26.389	35.611	35.167	32.667	25.722	28.722	44.111	34.667	22.667	35.944	31.111	40.167	37.333	22.111	36.889	29.944	46.389	33.278	28.222	31.111	32.333	31.444	32.833	40.667	22.944	36.778	29.778	34.611	37.389	20.889	33.389	34.167	23.833	38.889	31.722	45.056	31.000	34.944	35.722	27.222	31.056	24.611	44.889	27.611
	D	H	242.822	250.703	235.761	279.563	272.823	276.209	256.022	279.185	236.399	282.528	179.216	284.620	323.846	286.076	225.660	283.089	239.962	246.934	257.896	268.844	281.569	247.146	223.607	256.213	289.623	257.723	234.319	238.643	291.981	227.993	254.699	205.206	261.817	272.128	260.877	184.633	247.640	266.954	229.863	282.178	265.770	302.253	244.041	328.645	236.510	261.313	273.937	230.237	243.215	225.922
	M	N	0.611	0.667	0.667	0.833	0.444	0.611	0.611	0.722	0.500	0.667	0.667	0.667	0.556	0.611	0.500	0.500	0.833	0.611	0.333	0.722	0.778	0.889	0.667	0.611	0.944	0.833	0.611	0.556	0.778	0.778	0.833	0.667	0.667	0.500	0.944	0.722	0.778	0.889	0.944	0.722	0.778	0.889	0.556	0.611	0.722	0.611	0.667	0.833	0.667	0.444
	M	T	53.889	45.722	44.667	34.556	54.000	49.778	45.500	41.056	61.667	45.667	47.444	51.778	49.889	55.722	58.444	58.833	39.944	53.222	62.556	44.556	50.944	36.944	48.111	53.056	28.556	38.889	54.500	51.833	38.667	45.000	41.778	42.111	49.278	55.000	33.611	33.722	36.222	40.278	37.833	44.611	44.167	36.389	46.778	43.889	53.222	47.167	44.167	39.389	42.000	56.833
	M	H	321.301	294.623	284.437	249.231	302.087	264.005	340.197	325.831	326.454	306.851	241.852	350.267	300.767	301.438	294.497	305.675	253.801	270.048	309.989	305.437	285.618	279.461	271.748	307.652	286.040	282.433	322.824	298.939	267.305	292.807	282.524	280.953	351.974	280.886	253.419	231.290	260.119	296.789	314.722	299.667	312.455	272.729	293.098	262.568	349.317	295.969	260.028	292.423	299.006	328.870
	B	N	0.611	0.722	0.222	0.389	0.722	0.778	0.556	0.722	0.611	0.611	0.722	0.556	0.333	0.667	0.833	0.667	0.278	0.389	0.667	0.444	0.611	0.722	0.722	0.667	0.611	0.389	0.444	0.778	0.500	0.611	0.278	0.833	0.667	0.333	0.278	0.500	0.389	0.667	0.556	0.500	0.389	0.333	0.667	0.167	0.167	0.389	0.778	0.167	0.722	0.556
	B	T	46.778	38.722	66.222	62.889	43.056	48.722	51.611	49.000	58.444	58.056	47.944	45.611	63.444	46.722	48.056	43.056	64.389	57.722	38.778	52.833	54.722	54.667	47.000	51.333	43.833	59.444	52.833	41.889	56.111	42.944	66.444	33.611	53.556	62.389	67.222	68.944	56.833	44.500	54.000	46.389	58.944	69.500	41.556	71.889	68.167	63.500	45.667	73.111	56.333	56.000
	B	H	319.443	288.698	391.251	382.933	332.286	302.877	325.233	299.834	336.137	285.766	284.117	309.635	310.081	252.438	281.691	303.712	379.967	329.436	259.660	331.338	375.849	308.524	315.341	318.241	255.623	314.977	371.319	272.165	304.061	296.864	347.101	256.153	356.313	387.622	358.119	378.051	278.649	261.868	315.901	311.895	403.266	365.544	290.988	346.456	405.145	328.714	270.664	406.618	390.771	347.797
	I	N	0.778	0.667	0.944	0.833	0.722	0.667	0.833	0.778	0.889	0.889	0.778	0.889	0.833	0.833	0.889	0.944	0.833	0.778	0.833	0.889	0.944	0.944	0.778	0.889	0.833	0.944	0.833	0.944	0.944	0.889	0.944	0.833	0.667	0.889	0.889	0.778	0.889	0.833	0.889	0.833	0.944	0.944	0.722	0.833	0.833	0.944	0.667	0.667	0.833	0.944
	I	T	35.056	38.111	22.667	31.611	39.722	36.889	39.833	37.889	24.167	32.444	32.500	31.611	34.556	31.944	31.111	29.722	28.833	35.500	29.889	24.889	22.778	32.722	30.333	27.333	36.611	21.278	31.722	26.722	26.278	31.611	24.667	30.833	38.333	28.722	24.778	35.111	29.944	32.444	30.222	42.500	29.667	27.167	37.500	29.667	33.444	26.222	36.611	39.000	30.222	25.778
	I	H	235.512	255.226	251.746	213.725	264.978	263.834	184.116	211.556	242.199	219.520	237.989	206.363	260.899	243.382	237.241	215.144	214.359	219.805	275.432	204.463	206.035	224.097	218.575	211.299	230.488	197.190	212.128	195.664	197.059	241.862	216.422	248.489	252.183	220.055	200.373	197.388	254.241	224.900	235.294	241.007	177.515	219.972	221.393	209.874	243.223	262.979	253.520	230.499	271.683	228.898
	P	N	0.889	0.556	0.778	0.889	0.667	0.944	0.833	0.833	1.000	0.889	0.778	0.722	0.778	0.944	0.722	1.000	0.778	0.611	0.833	0.889	0.889	0.833	0.778	0.944	0.778	0.667	0.833	0.944	0.944	0.778	1.000	0.944	0.833	0.889	0.833	0.778	0.778	0.500	0.889	1.000	0.889	0.833	0.778	0.944	0.722	1.000	0.944	0.889	1.000	0.778
	P	T	37.056	53.556	40.444	26.167	46.944	30.111	32.167	35.444	36.333	35.222	36.444	37.833	38.556	28.833	39.667	35.722	46.611	45.278	35.833	29.944	36.111	30.667	37.167	28.778	36.444	42.944	30.444	30.500	32.389	36.389	29.111	32.667	35.000	35.167	50.500	44.222	38.556	49.944	44.278	23.056	33.611	27.722	45.333	34.889	40.722	26.611	28.444	31.389	22.000	38.444
	P	H	226.966	225.443	269.315	229.558	271.605	254.382	217.648	192.568	238.203	210.653	240.545	238.720	243.688	253.346	237.227	269.490	235.362	276.157	208.317	201.670	271.838	240.662	268.226	169.903	300.858	250.381	254.828	234.117	216.206	220.642	231.835	248.766	263.340	226.874	248.064	249.790	215.903	276.984	269.158	274.584	199.260	252.375	242.229	240.849	233.494	269.469	238.798	221.457	249.343	230.578
	T	N	0.889	0.778	0.722	0.889	0.889	0.722	0.944	0.833	0.667	0.667	0.722	0.833	0.833	0.944	0.667	0.889	1.000	0.889	0.778	0.889	0.667	0.833	0.722	0.944	0.833	0.833	0.833	0.611	0.611	0.944	0.778	0.944	0.778	0.667	0.667	0.833	0.722	0.667	0.722	0.833	0.667	0.889	0.722	0.833	0.778	0.722	0.722	0.667	0.944	0.833
	T	T	29.889	37.111	36.556	29.944	38.778	42.333	20.833	32.278	40.389	44.333	45.833	33.611	41.500	20.500	41.722	39.889	24.000	38.389	39.889	37.389	47.556	37.389	43.333	22.444	30.222	30.278	38.056	51.000	48.167	20.389	43.167	48.333	39.722	44.722	39.444	37.722	40.889	49.278	43.611	32.833	44.889	31.889	42.222	41.389	36.667	39.611	41.722	41.556	32.333	32.278
	T	H	257.624	271.770	242.243	254.210	276.604	276.022	268.315	193.022	292.011	287.156	302.496	287.750	251.992	271.736	315.581	267.626	267.434	213.436	250.532	227.400	276.698	241.002	255.894	295.468	284.828	232.200	240.771	223.076	254.899	231.302	267.118	274.373	283.493	273.547	259.090	272.281	250.625	275.795	268.430	255.648	216.682	261.515	267.514	277.948	281.837	218.848	244.671	285.680	230.263	244.995
t9r3	D	N	0.667	0.667	0.889	0.667	0.556	0.667	0.333	0.667	0.778	0.778	0.556	0.444	0.667	0.556	0.889	0.889	0.778	0.778	0.667	0.667	0.667	0.556	1.000	0.444	0.889	0.667	0.556	0.778	0.556	0.778	0.889	1.000	0.778	0.778	0.667	0.667	0.333	0.556	0.667	0.778	0.667	0.778	0.778	0.667	0.667	0.556	0.778	0.667	0.667	0.556
	D	T	36.778	32.222	34.222	37.889	46.667	36.667	58.778	46.667	31.556	29.778	42.444	52.333	36.778	48.444	22.889	21.667	36.000	33.222	32.222	41.444	38.778	43.222	26.333	56.667	27.889	42.333	44.778	28.889	42.444	32.667	24.778	14.667	27.889	32.778	36.667	37.111	61.444	49.778	40.556	32.000	36.333	35.667	32.556	46.556	41.778	42.889	34.444	35.778	43.667	45.333
	D	H	191.379	159.462	194.070	125.460	167.950	118.351	149.441	184.153	190.207	150.348	185.774	172.225	191.983	223.402	151.674	141.272	171.179	212.035	164.644	178.078	175.177	187.077	183.298	154.967	157.028	207.666	178.791	192.277	171.516	162.784	187.083	206.207	184.404	187.816	192.594	173.154	138.827	192.185	148.263	129.740	100.652	134.664	188.057	212.994	167.096	190.022	184.975	199.195	205.625	157.513
	M	N	0.778	0.667	0.667	0.667	0.667	0.667	0.667	0.667	0.667	0.889	0.667	0.333	0.444	0.556	0.667	0.444	0.667	0.778	0.444	0.444	0.556	0.556	0.778	0.778	0.778	0.778	0.667	0.667	0.667	0.778	0.889	0.556	0.556	0.667	0.889	0.667	0.556	0.667	0.444	0.778	0.667	0.778	0.444	0.778	0.778	0.667	0.778	0.778	0.556	0.444
	M	T	52.556	47.667	44.000	51.333	43.222	47.333	43.889	43.778	43.778	36.556	47.556	58.222	56.556	50.333	52.778	50.778	42.111	41.333	54.889	63.889	51.111	49.556	51.778	43.000	45.556	33.667	41.333	48.778	42.556	49.222	40.111	49.667	44.556	50.444	36.000	42.556	51.000	45.111	62.333	46.444	54.556	36.889	54.889	41.222	38.222	49.111	45.778	44.778	46.000	56.333
	M	H	183.842	150.550	162.211	208.625	194.110	168.674	176.166	150.059	164.660	225.840	212.396	187.639	196.607	199.746	238.685	173.178	165.045	162.429	178.781	246.355	214.227	227.722	213.193	190.022	167.074	162.221	226.787	224.242	208.971	225.083	207.175	184.074	212.338	155.940	173.914	214.617	211.951	152.652	230.402	210.331	209.048	187.957	182.463	222.493	171.054	213.945	172.098	175.254	222.113	162.469
	B	N	0.667	0.556	0.333	0.333	0.333	0.222	0.667	0.444	0.556	0.333	0.111	0.444	0.333	0.667	0.333	0.667	0.778	0.556	0.667	0.333	0.333	0.778	0.667	0.444	0.556	0.889	0.333	0.333	0.222	0.333	0.333	0.556	0.222	0.556	0.778	0.444	0.333	0.667	0.444	0.333	0.556	0.556	0.556	0.444	0.778	0.667	0.778	0.333	0.333	0.778
	B	T	40.778	53.556	68.889	66.889	60.667	70.444	44.667	56.333	56.889	71.778	72.889	60.778	57.000	43.111	59.111	45.000	53.222	51.111	59.111	56.333	70.667	54.556	52.556	55.889	55.889	45.111	58.556	59.889	67.778	67.778	68.444	49.667	70.667	52.333	41.000	48.444	70.000	50.556	64.667	61.667	54.333	52.889	51.556	59.000	44.444	54.000	37.778	63.333	64.111	52.667
	B	H	193.192	211.679	253.834	183.525	232.795	249.532	179.616	222.848	254.814	221.207	267.051	234.092	227.629	232.455	273.174	200.964	212.452	235.412	234.127	253.712	183.605	199.580	226.111	172.893	229.164	202.485	229.037	257.622	193.451	254.527	254.797	230.586	243.604	205.371	181.350	167.353	245.961	221.163	274.499	261.749	263.831	235.123	219.493	252.567	229.492	229.424	199.201	239.055	178.323	213.206
	I	N	0.889	0.778	0.778	0.889	0.778	0.889	0.889	0.889	0.778	0.778	0.889	0.778	0.889	0.778	1.000	0.889	0.556	0.889	0.889	0.778	0.667	0.889	1.000	0.889	0.556	1.000	1.000	0.889	0.889	0.667	0.889	0.778	0.556	0.778	1.000	0.667	0.778	0.889	0.889	0.667	0.778	0.889	1.000	0.889	1.000	0.889	0.889	0.889	0.667	0.556
	I	T	44.778	39.111	43.333	42.111	30.222	37.889	42.889	33.111	36.000	41.222	40.111	35.333	48.667	39.778	30.778	30.667	45.889	32.778	36.778	48.333	44.889	21.556	28.111	28.333	44.667	22.444	34.444	37.444	32.222	37.778	30.778	34.444	53.000	47.889	28.778	42.778	43.778	21.000	33.444	48.333	38.000	31.667	24.889	39.889	24.333	24.667	23.222	31.444	46.333	44.000
	I	H	160.400	161.978	152.161	171.745	194.897	149.004	162.825	143.347	170.092	179.939	151.718	183.025	169.426	157.828	169.985	169.444	215.199	196.292	179.585	158.909	141.369	171.847	138.158	153.966	161.619	155.074	161.925	148.790	217.422	205.392	148.807	170.870	221.400	184.397	165.519	173.138	163.902	155.580	157.622	142.026	143.146	161.401	184.273	139.295	183.338	136.326	149.328	127.021	155.434	162.610
	P	N	0.667	0.667	1.000	0.778	0.556	0.889	0.889	0.778	0.889	0.444	0.778	0.667	1.000	0.778	1.000	0.889	0.556	0.778	0.667	0.889	0.667	1.000	0.778	0.667	0.778	0.889	0.778	0.778	0.556	0.778	0.778	0.889	0.889	0.667	0.667	0.667	0.556	0.667	0.778	0.778	0.444	0.889	0.667	0.556	0.889	0.667	0.889	0.778	0.778	0.778
	P	T	38.333	35.444	27.556	42.444	51.222	36.333	31.111	43.444	45.000	56.778	35.889	41.778	27.222	36.111	34.778	35.778	43.889	36.889	42.222	37.222	50.667	25.444	49.889	47.778	36.000	38.000	36.111	42.556	44.778	41.111	38.667	29.444	25.000	45.778	45.667	42.778	45.444	42.000	33.111	36.889	55.889	36.000	36.222	52.333	39.222	41.778	34.667	37.000	36.000	38.889
	P	H	184.293	184.939	210.327	149.042	160.368	133.983	191.662	182.470	173.393	177.576	169.856	146.108	130.391	186.121	189.534	169.646	167.488	146.510	176.120	133.398	189.641	134.983	190.439	145.204	187.885	132.010	187.480	134.698	148.729	164.131	143.730	166.400	138.812	191.022	194.739	142.898	139.185	149.111	180.009	140.854	144.437	145.902	192.970	155.133	144.469	195.980	178.794	155.558	122.868	186.060
	T	N	1.000	0.556	0.556	1.000	0.667	0.667	0.667	0.667	0.667	1.000	0.778	0.667	1.000	0.778	0.667	0.889	0.778	0.889	0.778	0.667	0.889	0.667	0.667	0.778	1.000	0.667	1.000	1.000	0.556	0.444	0.667	0.778	0.556	0.889	0.667	0.667	0.556	1.000	0.667	0.778	0.667	0.889	1.000	0.667	0.556	1.000	0.778	0.778	0.667	0.667
	T	T	38.667	41.111	51.444	32.111	52.667	52.333	51.222	54.778	43.000	30.667	44.667	55.444	38.778	36.667	49.000	47.444	38.667	30.667	33.667	47.000	32.444	41.778	37.444	30.222	44.000	47.222	36.778	41.556	50.444	61.222	46.778	39.333	46.444	36.222	51.556	36.000	56.333	28.556	37.556	43.333	54.000	27.222	27.556	42.667	51.444	29.778	31.889	38.667	43.222	55.111
	T	H	200.905	210.693	177.442	145.012	157.412	175.653	190.727	187.185	160.886	180.666	188.326	191.167	188.549	217.721	204.266	159.019	162.385	203.429	169.492	199.936	170.419	143.481	204.921	132.616	186.617	171.480	190.685	186.639	193.089	203.253	150.217	179.557	192.242	186.218	219.495	156.015	170.318	162.280	169.351	179.649	184.062	168.863	135.616	190.158	181.101	184.991	184.896	197.665	158.680	207.374

M：D 为 DDPG，M 为 MHA-PPO，B 为 Boust，I 为本文方法，P 为 MAPPO，T 为 PTR；K：N 为目标发现比例，T 为目标发现时间，H 为回合平均信息熵。

表 D.2: 第四章观测聚合多场景测试原始数据

Cfg	M	K	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49
A10S5	A	D	59.661	56.886	66.527	62.788	58.366	60.833	68.297	59.668	69.138	49.507	60.378	61.972	62.504	53.240	56.113	60.255	66.994	60.695	57.347	60.320	65.627	66.352	66.066	56.773	59.527	58.774	56.044	54.318	63.876	62.353	64.124	65.558	71.464	66.919	66.963	55.734	69.407	57.251	65.846	54.194	56.987	66.169	59.579	62.635	55.199	51.075	59.122	62.822	68.586	61.766
	A	U	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.600	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.900	0.800	1.000	0.900	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	A	S	0.800	1.000	1.000	1.000	1.000	1.000	1.000	0.800	1.000	0.600	1.000	1.000	1.000	0.600	0.800	1.000	1.000	0.800	1.000	1.000	1.000	1.000	1.000	0.600	0.800	0.600	0.400	0.600	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.800	1.000	1.000	1.000	0.400	1.000	1.000	1.000	1.000
	A	R	104.001	113.813	144.746	75.421	47.120	16.538	85.043	23.318	112.751	25.446	75.150	122.558	2.698	40.592	44.257	44.611	24.165	44.948	22.941	62.509	82.497	127.945	136.934	66.102	10.405	11.977	91.712	-4.912	51.512	119.498	97.282	146.927	104.352	63.904	82.953	23.413	54.902	84.602	139.331	27.436	17.128	160.753	53.464	79.259	31.655	25.581	85.159	42.344	94.308	42.363
	A	W	0.000	0.500	0.667	0.750	0.800	0.833	0.857	0.750	0.778	0.700	0.727	0.750	0.769	0.714	0.667	0.688	0.706	0.667	0.684	0.700	0.714	0.727	0.739	0.708	0.680	0.654	0.630	0.607	0.621	0.633	0.645	0.656	0.667	0.676	0.686	0.694	0.703	0.711	0.718	0.725	0.732	0.714	0.721	0.705	0.711	0.696	0.702	0.708	0.714	0.720
	F	D	48.533	59.244	36.933	48.289	55.795	50.077	47.060	61.883	55.280	59.318	50.103	50.300	53.302	55.436	54.938	51.094	53.046	41.722	47.317	47.554	55.271	56.725	59.872	56.439	59.685	53.526	47.022	58.874	47.898	56.480	62.741	49.479	55.045	59.942	53.727	59.841	56.413	51.531	50.606	55.775	56.495	59.222	58.066	49.974	57.388	50.971	52.676	57.712	43.640	51.719
	F	U	1.000	1.000	0.500	1.000	1.000	1.000	0.800	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.700	1.000	1.000	0.800	1.000	0.700	0.900	1.000	1.000	1.000	1.000	0.800	1.000	1.000	1.000	0.800	0.900	0.900	1.000	1.000	1.000	1.000	1.000	1.000	0.700	1.000	1.000	1.000	1.000	0.900	0.800	1.000	0.900
	F	S	0.600	1.000	0.200	1.000	1.000	1.000	0.600	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.000	1.000	1.000	0.400	1.000	0.600	0.600	1.000	1.000	1.000	1.000	0.400	1.000	1.000	1.000	0.400	0.600	0.400	0.800	1.000	1.000	1.000	1.000	1.000	0.600	1.000	0.400	1.000	1.000	0.800	0.600	1.000	0.200
	F	R	-30.725	-20.553	-23.176	8.037	18.157	-55.170	12.533	-5.948	54.993	27.480	20.069	-17.521	-10.647	62.841	-15.775	28.834	-22.075	-133.282	8.569	4.493	-53.019	72.939	-18.272	-12.763	36.012	48.386	47.589	0.355	53.925	-28.003	102.400	-1.952	-91.085	-50.339	-43.606	27.697	-17.812	-26.351	-32.847	77.392	22.573	-18.697	30.610	21.717	3.049	11.980	61.406	-64.420	-35.818	-90.782
	F	W	0.000	0.500	0.333	0.500	0.600	0.667	0.571	0.625	0.667	0.700	0.636	0.667	0.692	0.714	0.733	0.750	0.765	0.722	0.737	0.750	0.714	0.727	0.696	0.667	0.680	0.692	0.704	0.714	0.690	0.700	0.710	0.719	0.697	0.676	0.657	0.639	0.649	0.658	0.667	0.675	0.683	0.667	0.674	0.659	0.667	0.674	0.660	0.646	0.653	0.640
	G	D	68.752	63.012	67.619	68.743	77.080	57.929	61.019	68.816	59.247	53.945	55.568	60.356	57.693	75.810	58.798	65.197	69.087	55.597	68.773	69.630	59.429	65.613	59.325	60.199	67.189	53.836	56.666	66.526	64.183	59.054	70.319	58.260	71.344	63.039	56.344	70.690	64.295	66.978	62.198	51.961	61.774	66.594	61.480	63.450	53.659	62.804	55.134	62.578	66.820	58.546
	G	U	1.000	1.000	1.000	1.000	1.000	0.900	1.000	1.000	1.000	0.800	0.900	0.800	0.700	1.000	0.800	1.000	1.000	0.700	1.000	1.000	1.000	1.000	0.800	1.000	1.000	1.000	0.800	1.000	1.000	1.000	1.000	0.900	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.800	0.800	1.000	1.000	0.900	1.000	1.000	0.700	1.000	1.000	1.000
	G	S	1.000	1.000	1.000	1.000	1.000	0.600	0.800	1.000	0.600	0.400	0.400	0.600	0.600	1.000	0.600	0.800	1.000	0.600	0.600	0.800	0.200	1.000	0.600	1.000	1.000	0.600	0.600	1.000	0.800	1.000	1.000	0.600	1.000	1.000	0.800	1.000	0.600	0.800	1.000	0.400	0.800	1.000	1.000	0.600	0.600	1.000	0.600	1.000	1.000	0.800
	G	R	157.587	27.365	133.141	90.446	84.937	13.260	88.337	30.107	-31.340	-9.440	92.499	60.905	55.546	137.272	12.477	95.122	67.206	-37.325	49.266	76.143	-1.883	70.810	15.211	32.845	66.701	41.803	5.476	181.701	50.208	51.679	96.556	-15.118	84.914	71.235	20.601	58.138	25.720	75.387	50.073	-51.560	38.367	83.184	40.504	36.627	19.854	59.101	-29.082	106.170	131.851	17.551
	G	W	1.000	1.000	1.000	1.000	1.000	0.833	0.714	0.750	0.667	0.600	0.545	0.500	0.462	0.500	0.467	0.438	0.471	0.444	0.421	0.400	0.381	0.409	0.391	0.417	0.440	0.423	0.407	0.429	0.414	0.433	0.452	0.438	0.455	0.441	0.429	0.444	0.432	0.421	0.436	0.425	0.415	0.429	0.442	0.432	0.422	0.435	0.426	0.417	0.429	0.420
	H	D	56.223	56.776	68.587	66.652	72.903	61.489	71.870	71.682	59.906	71.726	63.109	68.074	64.167	59.471	68.094	71.082	55.878	64.419	69.324	66.256	54.505	61.486	72.392	71.872	61.346	71.534	62.032	68.025	72.106	62.319	58.836	71.929	62.540	66.481	74.008	68.392	60.532	65.644	66.023	68.816	70.964	57.908	69.080	61.084	66.989	60.929	64.311	64.667	74.902	68.445
	H	U	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.900	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	H	S	0.600	1.000	0.800	1.000	1.000	0.600	1.000	1.000	1.000	1.000	0.600	1.000	1.000	1.000	1.000	1.000	0.800	0.800	1.000	1.000	1.000	0.800	1.000	1.000	0.600	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	H	R	19.309	7.587	46.370	96.111	146.606	33.441	129.834	73.969	89.132	72.915	99.869	61.421	67.089	57.836	78.194	140.922	6.183	74.822	99.364	21.799	10.450	60.948	76.432	146.718	7.982	101.496	34.488	140.093	50.352	17.344	80.630	113.811	40.292	135.160	101.919	89.934	125.866	82.228	95.019	105.063	56.367	34.840	6.299	78.823	35.910	112.864	86.779	75.485	106.447	117.530
	H	W	0.000	0.500	0.333	0.500	0.600	0.500	0.571	0.625	0.667	0.700	0.636	0.667	0.692	0.714	0.733	0.750	0.706	0.667	0.684	0.700	0.714	0.682	0.696	0.708	0.680	0.692	0.704	0.714	0.724	0.733	0.742	0.750	0.758	0.765	0.771	0.778	0.784	0.789	0.795	0.800	0.805	0.810	0.814	0.818	0.822	0.826	0.830	0.833	0.837	0.840
A12S6	A	D	—	61.758	62.036	65.467	57.442	69.628	67.299	57.519	60.578	57.916	68.311	67.759	63.865	62.425	61.979	66.304	62.943	66.024	76.765	59.660	69.116	59.772	65.141	59.177	62.650	63.317	66.867	66.049	68.239	62.342	65.178	59.093	65.573	60.896	67.884	55.319	58.478	72.450	68.862	66.804	62.566	64.399	62.138	71.250	62.782	68.062	66.363	57.497	60.567	67.728
	A	U	—	1.000	1.000	1.000	1.000	1.000	1.000	0.917	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.917	0.917	1.000	1.000	1.000	1.000	0.917	1.000	1.000	1.000	1.000	1.000	0.917	1.000	0.917
	A	S	—	1.000	1.000	1.000	1.000	1.000	1.000	0.667	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.833	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.833	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.667	0.500	1.000	1.000	1.000	1.000	0.667	1.000	1.000	1.000	1.000	1.000	0.667	0.667	1.000
	A	R	—	116.864	34.128	101.969	46.032	112.944	173.567	6.974	55.066	21.895	124.930	145.295	79.389	60.421	-53.304	25.225	41.011	209.089	128.097	49.190	90.752	3.250	78.844	4.304	-0.829	-20.937	136.407	7.362	42.331	71.078	58.971	17.723	92.772	52.873	65.517	-33.362	31.727	120.222	28.629	51.619	6.442	-1.521	33.413	127.175	53.527	66.263	84.540	27.569	15.481	41.022
	A	W	—	1.000	1.000	1.000	1.000	1.000	1.000	0.875	0.889	0.900	0.909	0.917	0.923	0.929	0.933	0.938	0.882	0.889	0.895	0.900	0.905	0.909	0.913	0.917	0.880	0.885	0.889	0.893	0.897	0.900	0.903	0.906	0.909	0.912	0.914	0.889	0.865	0.868	0.872	0.875	0.878	0.857	0.860	0.841	0.844	0.848	0.851	0.833	0.816	0.800
	F	D	54.301	56.878	53.579	52.813	52.731	51.563	59.031	59.442	55.836	52.454	56.284	53.476	63.644	59.258	57.205	48.737	57.411	61.154	48.160	60.578	55.750	51.500	49.694	55.538	58.205	54.228	47.485	54.076	46.806	51.398	55.843	53.568	52.216	58.300	52.321	59.147	44.806	58.826	53.620	53.330	53.291	50.045	55.359	48.104	53.522	55.214	53.021	49.792	53.898	50.813
	F	U	1.000	0.833	1.000	1.000	1.000	1.000	0.917	0.833	1.000	1.000	0.750	1.000	1.000	0.833	1.000	1.000	1.000	0.833	0.917	1.000	1.000	1.000	1.000	1.000	0.917	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.833	1.000	1.000	0.833	1.000	1.000	1.000	1.000	0.833	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	F	S	1.000	0.500	1.000	1.000	1.000	1.000	0.667	0.667	1.000	1.000	0.667	1.000	0.667	0.667	1.000	1.000	1.000	0.667	0.667	0.500	1.000	1.000	1.000	1.000	0.667	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.667	1.000	1.000	1.000	1.000	0.500	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	F	R	-1.120	-17.062	63.144	-23.827	-26.187	-63.010	13.260	65.060	76.879	-22.035	63.063	16.138	17.322	48.996	86.324	-9.488	-127.066	111.671	-56.662	18.249	16.994	19.838	-31.833	-37.867	-51.874	11.569	-32.489	18.759	-59.367	-67.121	36.344	-54.191	-81.576	-7.380	-16.922	45.983	-88.290	29.189	-16.927	-72.604	-5.030	-94.864	-74.004	11.894	12.871	16.733	-4.575	0.240	-102.561	-5.210
	F	W	1.000	0.500	0.667	0.750	0.800	0.833	0.714	0.625	0.667	0.700	0.636	0.667	0.615	0.571	0.600	0.625	0.647	0.611	0.579	0.550	0.571	0.591	0.609	0.625	0.600	0.615	0.630	0.643	0.655	0.667	0.677	0.688	0.667	0.676	0.686	0.667	0.676	0.684	0.692	0.700	0.683	0.690	0.698	0.705	0.711	0.717	0.723	0.729	0.735	0.740
	G	D	64.226	66.885	50.453	64.174	61.133	68.147	72.254	64.859	66.777	70.757	60.134	66.736	66.432	67.668	69.514	55.972	69.747	51.494	57.758	57.274	64.126	59.984	68.280	62.608	51.476	72.422	64.727	65.699	65.123	64.359	73.956	65.430	59.785	62.446	68.820	63.709	63.563	67.356	64.674	68.244	63.516	57.873	60.591	63.174	62.285	73.338	65.256	73.798	70.756	68.122
	G	U	1.000	1.000	0.833	1.000	0.917	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.833	1.000	1.000	1.000	1.000	1.000	1.000	0.750	1.000	1.000	1.000	1.000	1.000	1.000	0.917	1.000	0.917	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	G	S	1.000	1.000	0.500	1.000	0.833	1.000	0.833	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.667	1.000	0.500	1.000	1.000	1.000	1.000	1.000	1.000	0.500	1.000	1.000	1.000	0.667	1.000	1.000	0.833	0.667	0.667	1.000	1.000	1.000	1.000	0.833	1.000	0.667	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	G	R	39.757	58.485	-21.895	115.597	-20.708	149.770	182.216	48.111	84.068	132.042	57.073	166.000	126.415	82.449	85.738	30.167	128.956	-68.011	56.809	12.401	87.736	127.438	106.231	42.178	-92.717	101.641	31.495	80.989	95.072	31.238	117.103	67.834	-1.504	19.784	73.603	120.183	16.588	91.987	76.756	94.949	75.316	87.571	47.620	81.239	50.183	138.770	67.410	71.559	68.477	35.337
	G	W	1.000	1.000	0.667	0.750	0.600	0.667	0.571	0.625	0.667	0.700	0.727	0.750	0.769	0.786	0.800	0.750	0.765	0.722	0.737	0.750	0.762	0.773	0.783	0.792	0.760	0.769	0.778	0.786	0.759	0.767	0.774	0.750	0.727	0.706	0.714	0.722	0.730	0.737	0.718	0.725	0.707	0.714	0.721	0.727	0.733	0.739	0.745	0.750	0.755	0.760
	H	D	66.914	65.732	63.958	66.518	73.587	64.404	66.077	72.017	57.057	64.383	64.993	66.657	65.836	58.329	63.859	60.285	69.145	68.501	59.075	67.807	61.169	59.511	62.695	63.945	70.506	63.808	59.140	72.976	70.987	69.632	58.362	67.096	68.748	66.035	69.100	70.590	60.198	61.717	62.454	63.071	71.279	69.041	67.717	69.502	73.629	70.451	64.458	62.980	66.675	67.345
	H	U	1.000	1.000	1.000	1.000	1.000	1.000	0.917	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.917	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	H	S	1.000	1.000	1.000	1.000	1.000	1.000	0.500	1.000	1.000	1.000	1.000	1.000	1.000	0.833	1.000	1.000	1.000	1.000	0.667	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.667	1.000	1.000	0.833	1.000	1.000	1.000	1.000	1.000	0.667	0.833	1.000	1.000	1.000	1.000	1.000	1.000	0.667	1.000	1.000
	H	R	90.767	6.411	79.108	96.159	211.181	60.400	7.600	165.775	21.284	75.569	72.732	38.767	95.294	33.511	93.682	2.326	102.465	116.098	53.751	127.467	86.200	123.790	69.316	108.419	114.138	93.881	9.044	76.408	99.616	86.082	90.418	138.537	134.256	69.984	124.361	86.020	26.689	91.130	118.620	87.672	104.306	92.103	109.964	143.027	134.413	111.036	76.805	11.457	124.598	88.561
	H	W	1.000	1.000	1.000	1.000	1.000	1.000	0.857	0.875	0.778	0.800	0.818	0.833	0.846	0.786	0.800	0.812	0.824	0.833	0.789	0.800	0.810	0.818	0.826	0.833	0.840	0.846	0.852	0.857	0.862	0.867	0.839	0.844	0.848	0.824	0.829	0.833	0.838	0.842	0.846	0.825	0.805	0.810	0.814	0.818	0.822	0.826	0.830	0.812	0.816	0.820
A6S3	A	D	57.383	53.330	68.377	55.831	56.359	73.010	66.622	55.480	77.669	63.516	66.251	54.075	53.794	61.698	64.100	64.534	68.396	64.519	50.224	53.316	68.549	64.269	50.177	65.156	71.150	63.897	59.046	55.098	67.464	61.496	62.269	69.351	52.692	67.835	47.564	67.384	58.275	51.298	72.034	46.170	73.716	66.052	73.047	47.359	48.927	49.526	44.099	71.339	56.903	41.745
	A	U	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.667	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.833	1.000	1.000	1.000	0.833	1.000	1.000	0.500	1.000	1.000	0.667
	A	S	0.333	1.000	1.000	1.000	0.667	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.667	0.667	1.000	1.000	1.000	0.333	1.000	1.000	1.000	1.000	1.000	0.333	1.000	0.000	1.000	1.000	1.000	1.000	0.000	1.000	1.000	1.000	0.667	1.000	0.667	0.667	1.000	1.000	0.667
	A	R	-12.288	75.906	12.699	62.174	36.031	75.976	70.143	35.660	62.412	88.973	132.027	67.403	17.071	33.145	60.162	79.108	62.630	35.489	25.476	59.757	94.835	46.050	30.557	72.649	66.851	121.038	60.385	81.838	58.166	80.671	80.251	57.093	44.833	25.537	-5.590	96.861	101.377	-35.368	93.692	63.490	23.680	85.339	66.003	29.032	32.427	-8.489	-15.142	79.945	72.293	-7.880
	A	W	0.000	0.500	0.667	0.750	0.600	0.667	0.714	0.750	0.778	0.800	0.818	0.833	0.846	0.857	0.867	0.875	0.882	0.889	0.895	0.900	0.905	0.864	0.826	0.833	0.840	0.846	0.815	0.821	0.828	0.833	0.839	0.844	0.818	0.824	0.800	0.806	0.811	0.816	0.821	0.800	0.805	0.810	0.814	0.795	0.800	0.783	0.766	0.771	0.755	0.740
	F	D	45.710	43.656	52.148	52.690	53.828	41.436	51.495	62.669	56.067	42.221	43.993	60.446	50.202	45.492	51.380	43.733	53.028	53.049	47.534	63.286	47.336	39.061	56.203	54.497	47.920	40.121	52.691	60.212	50.213	59.179	55.844	57.809	55.534	41.950	53.811	54.992	50.869	49.585	59.514	54.136	45.682	53.639	60.180	41.535	62.498	48.711	49.187	41.512	52.457	67.086
	F	U	0.833	1.000	1.000	1.000	0.833	0.833	0.833	1.000	1.000	0.833	0.833	1.000	1.000	1.000	1.000	0.833	0.833	1.000	1.000	0.833	1.000	0.833	1.000	0.833	1.000	1.000	1.000	0.833	0.833	1.000	1.000	0.833	0.667	0.833	1.000	0.667	1.000	1.000	0.833	1.000	1.000	1.000	0.833	0.667	1.000	1.000	1.000	0.833	1.000	1.000
	F	S	0.333	0.333	1.000	1.000	0.667	0.667	0.667	0.667	1.000	0.667	0.667	1.000	1.000	1.000	1.000	0.667	0.333	1.000	0.667	0.667	0.667	0.000	0.333	1.000	1.000	1.000	1.000	0.667	0.667	0.667	1.000	0.667	0.667	0.333	1.000	0.667	1.000	1.000	0.667	1.000	1.000	1.000	0.667	0.333	1.000	1.000	1.000	0.333	1.000	1.000
	F	R	-8.575	41.533	62.312	43.152	41.616	-78.141	35.364	81.577	67.779	-47.649	36.504	8.963	20.591	-40.741	25.368	24.641	12.506	63.233	21.440	8.250	-75.945	-9.404	53.699	21.380	49.189	-16.224	-2.729	23.017	-19.248	11.194	40.748	16.921	-5.725	1.916	23.902	14.316	65.569	7.274	-0.741	55.236	-16.747	16.053	57.626	-10.389	91.911	49.125	15.697	-65.305	11.113	51.255
	F	W	0.000	0.000	0.333	0.500	0.400	0.333	0.286	0.250	0.333	0.300	0.273	0.333	0.385	0.429	0.467	0.438	0.412	0.444	0.421	0.400	0.381	0.364	0.348	0.333	0.360	0.385	0.407	0.393	0.379	0.367	0.387	0.375	0.364	0.353	0.371	0.361	0.378	0.395	0.385	0.400	0.415	0.429	0.419	0.409	0.422	0.435	0.447	0.438	0.449	0.460
	G	D	56.586	54.067	43.318	55.505	47.437	56.276	66.680	55.215	60.275	65.504	49.840	73.057	49.417	43.625	76.499	49.940	54.196	47.399	59.556	66.630	74.619	57.324	42.401	61.640	51.756	62.048	51.319	66.088	46.015	48.665	58.047	60.589	63.755	71.980	52.177	68.767	57.116	60.187	60.269	57.935	59.489	38.701	50.666	49.345	47.032	57.953	56.023	66.781	42.043	62.935
	G	U	1.000	0.667	0.667	1.000	0.667	0.833	1.000	1.000	1.000	1.000	0.833	1.000	1.000	1.000	1.000	0.833	1.000	0.667	1.000	1.000	1.000	1.000	0.667	1.000	1.000	1.000	0.833	1.000	0.667	0.833	0.667	1.000	0.833	1.000	0.833	1.000	1.000	1.000	1.000	1.000	0.833	0.833	1.000	1.000	1.000	0.833	0.833	0.833	0.833	0.833
	G	S	1.000	0.333	0.333	0.333	0.333	0.667	1.000	1.000	1.000	0.667	0.333	1.000	1.000	0.667	1.000	0.333	0.333	0.333	1.000	0.667	1.000	0.667	0.000	1.000	0.667	1.000	0.333	1.000	0.667	0.333	0.667	0.667	0.667	1.000	0.333	1.000	1.000	0.333	1.000	1.000	0.667	0.000	1.000	1.000	0.667	0.333	0.333	0.667	0.000	0.667
	G	R	42.811	-49.284	-20.266	76.795	-7.438	31.750	91.776	90.582	71.900	100.507	-14.594	102.434	11.435	-15.052	48.595	-21.435	2.137	-17.690	44.027	60.103	63.419	57.343	28.286	71.385	-16.166	27.009	11.484	61.619	-14.528	9.264	22.620	36.349	8.362	44.549	9.522	116.660	-29.551	85.039	25.640	52.001	104.079	-57.714	42.587	63.179	6.685	-6.163	3.156	33.466	6.318	14.020
	G	W	1.000	0.500	0.333	0.250	0.200	0.167	0.286	0.375	0.444	0.400	0.364	0.417	0.462	0.429	0.467	0.438	0.412	0.389	0.421	0.400	0.429	0.409	0.391	0.417	0.400	0.423	0.407	0.393	0.379	0.367	0.355	0.344	0.333	0.324	0.314	0.333	0.351	0.342	0.359	0.375	0.366	0.357	0.372	0.386	0.378	0.370	0.362	0.354	0.347	0.340
	H	D	59.317	70.810	53.520	50.463	66.429	65.836	58.647	68.449	60.889	68.345	71.946	68.922	67.650	49.111	61.539	60.072	60.587	63.422	60.530	55.798	69.872	58.231	74.346	64.069	63.088	57.301	57.220	59.235	60.980	69.253	75.716	59.517	58.054	47.040	59.282	63.952	62.360	59.644	60.945	67.516	65.861	65.904	67.441	70.255	72.679	59.239	65.510	58.310	48.770	73.960
	H	U	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.833	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	H	S	1.000	1.000	0.667	0.667	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.667	1.000	1.000	0.333	1.000	1.000	1.000	1.000	1.000	0.333	1.000	0.667	1.000	1.000	1.000	1.000	0.667	1.000	1.000	1.000	0.667	1.000	0.667	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	H	R	17.081	45.290	54.785	39.215	37.296	57.018	27.949	99.341	12.967	97.836	70.351	58.555	6.440	43.740	63.280	-2.629	27.707	17.562	46.360	32.941	59.327	40.496	109.950	86.782	35.242	-24.868	62.284	-48.429	20.448	131.825	49.651	21.575	49.692	43.326	49.742	63.825	44.696	49.846	34.866	54.741	34.680	79.970	77.209	49.319	67.392	68.285	82.052	47.046	1.564	94.291
	H	W	1.000	0.500	0.333	0.250	0.400	0.500	0.571	0.625	0.667	0.700	0.727	0.750	0.692	0.714	0.733	0.688	0.647	0.667	0.684	0.650	0.667	0.682	0.696	0.708	0.720	0.692	0.704	0.679	0.690	0.700	0.710	0.719	0.697	0.706	0.714	0.722	0.703	0.711	0.692	0.700	0.707	0.714	0.721	0.727	0.733	0.739	0.745	0.750	0.755	0.760
A8S4	A	D	59.967	58.620	60.059	66.708	53.884	62.146	59.832	50.038	52.530	56.652	62.057	54.490	60.625	67.442	53.665	60.017	66.035	62.663	54.798	57.787	63.186	59.273	58.334	67.114	53.914	66.348	60.806	53.481	62.149	59.248	62.703	56.257	56.563	68.840	50.568	64.672	59.409	63.285	47.955	54.738	61.025	54.720	60.523	63.953	61.573	61.233	63.669	61.884	61.479	67.739
	A	U	1.000	1.000	1.000	1.000	0.875	1.000	1.000	0.625	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	A	S	1.000	1.000	1.000	1.000	0.750	1.000	1.000	0.750	1.000	1.000	1.000	1.000	0.750	0.750	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.750	0.750	1.000	1.000	1.000	1.000	1.000	1.000	0.500	0.750	1.000	1.000	1.000	0.750	1.000	0.500	1.000	1.000	0.750	1.000	0.750	1.000	1.000	1.000	1.000	1.000	1.000
	A	R	35.002	47.112	49.577	114.198	-27.214	98.025	24.564	-13.630	77.418	37.337	95.437	24.451	30.769	37.360	47.766	24.773	71.632	57.899	15.507	40.560	54.756	23.234	14.091	67.495	12.380	88.871	100.546	49.136	81.218	15.234	33.159	69.837	20.669	72.363	53.732	91.885	3.060	135.375	31.912	58.528	100.236	-14.506	90.185	56.301	40.724	110.121	42.647	30.418	46.408	114.667
	A	W	1.000	1.000	1.000	1.000	0.800	0.833	0.857	0.750	0.778	0.800	0.818	0.833	0.769	0.714	0.667	0.688	0.706	0.722	0.737	0.750	0.762	0.773	0.783	0.750	0.720	0.731	0.741	0.750	0.759	0.767	0.774	0.750	0.727	0.735	0.743	0.750	0.730	0.737	0.718	0.725	0.732	0.714	0.721	0.705	0.711	0.717	0.723	0.729	0.735	0.740
	F	D	53.040	46.988	37.001	47.669	51.213	51.439	57.649	43.895	40.153	47.319	49.751	52.632	53.516	45.779	44.347	54.148	62.545	51.986	39.171	56.616	57.726	39.629	49.106	58.229	43.501	38.976	49.145	44.191	54.568	54.413	50.247	52.818	41.893	48.947	52.940	54.695	33.129	56.440	38.346	46.149	51.356	54.478	52.918	52.408	52.289	36.516	39.442	45.321	47.129	43.451
	F	U	1.000	1.000	0.625	1.000	0.875	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	0.875	0.875	1.000	1.000	1.000	0.750	1.000	0.875	1.000	1.000	1.000	0.625	0.875	1.000	1.000	1.000	1.000	1.000	0.750	0.875	1.000	1.000	0.875	0.625	1.000	0.875	1.000	1.000	1.000	0.875	0.875	1.000	1.000	0.875	0.750	1.000	0.875
	F	S	0.750	1.000	0.500	1.000	0.500	1.000	1.000	1.000	0.500	1.000	0.500	1.000	0.500	1.000	0.500	1.000	0.750	1.000	0.250	1.000	0.750	0.750	0.750	1.000	0.500	0.000	1.000	0.750	1.000	1.000	1.000	0.500	0.500	1.000	1.000	0.500	0.000	1.000	0.500	1.000	1.000	1.000	0.500	0.500	1.000	0.500	0.500	0.500	1.000	0.000
	F	R	45.125	13.133	-21.530	4.568	-39.428	-19.648	50.078	6.191	-22.527	-16.530	47.582	14.019	42.431	-3.517	30.827	36.483	93.836	7.118	-28.076	53.799	81.905	2.917	14.613	94.428	-96.213	22.282	4.269	41.029	75.058	12.904	-11.270	12.699	-11.510	19.994	52.240	9.144	-141.224	56.596	21.762	10.262	12.414	-43.193	34.566	-32.786	45.562	-7.841	-27.832	-1.997	0.997	-11.461
	F	W	0.000	0.500	0.333	0.500	0.400	0.500	0.571	0.625	0.556	0.600	0.545	0.583	0.538	0.500	0.467	0.500	0.471	0.500	0.474	0.500	0.476	0.455	0.435	0.458	0.440	0.423	0.444	0.429	0.448	0.467	0.484	0.469	0.455	0.471	0.486	0.472	0.459	0.474	0.462	0.475	0.488	0.500	0.488	0.477	0.489	0.478	0.468	0.458	0.469	0.460
	G	D	65.927	59.511	53.009	53.889	46.703	47.054	58.647	68.616	66.949	51.492	51.226	46.830	54.446	50.577	61.221	62.392	52.483	53.586	60.234	58.996	75.716	53.236	70.718	59.216	66.699	47.808	62.307	71.323	50.023	61.738	62.031	57.153	60.787	48.759	66.116	58.771	51.002	52.544	70.034	47.923	54.960	61.076	63.310	44.775	59.206	63.997	59.417	74.391	77.086	60.052
	G	U	1.000	1.000	0.875	0.875	0.750	0.750	1.000	1.000	1.000	1.000	0.875	0.750	1.000	0.875	1.000	1.000	0.625	1.000	1.000	1.000	1.000	0.875	1.000	0.875	1.000	0.625	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	0.750	0.875	0.625	0.875	1.000	1.000	0.625	0.875	1.000	1.000	1.000	1.000	0.875
	G	S	1.000	1.000	0.500	0.250	0.250	0.000	1.000	1.000	1.000	0.000	0.250	0.000	1.000	0.500	0.750	0.500	0.500	0.500	1.000	0.750	1.000	0.500	0.750	0.500	1.000	0.250	1.000	1.000	0.500	1.000	1.000	0.750	0.750	0.750	1.000	0.750	0.250	0.250	1.000	0.250	0.500	0.500	0.750	0.500	0.750	0.750	0.750	1.000	1.000	0.750
	G	R	36.689	28.788	8.535	3.567	-47.938	-37.607	60.753	60.181	35.934	-4.598	5.046	-13.880	42.421	-51.550	42.748	59.233	-71.151	26.316	78.499	41.018	88.705	37.526	106.202	-15.145	174.509	-55.660	78.595	47.935	25.896	38.568	66.633	26.053	77.763	16.553	38.060	37.071	10.286	-11.051	142.976	-17.756	17.991	14.675	83.158	-14.383	-5.609	91.173	20.957	131.279	113.160	85.628
	G	W	1.000	1.000	0.667	0.500	0.400	0.333	0.429	0.500	0.556	0.500	0.455	0.417	0.462	0.429	0.400	0.375	0.353	0.333	0.368	0.350	0.381	0.364	0.348	0.333	0.360	0.346	0.370	0.393	0.379	0.400	0.419	0.406	0.394	0.382	0.400	0.389	0.378	0.368	0.359	0.350	0.341	0.333	0.326	0.318	0.311	0.304	0.298	0.312	0.327	0.320
	H	D	63.417	57.153	56.436	58.767	63.317	60.520	54.963	66.251	67.463	56.664	67.181	63.871	65.865	77.152	70.210	63.145	68.253	59.074	59.187	70.497	68.458	72.710	56.060	74.697	58.907	66.679	62.108	68.312	66.879	63.848	68.364	57.583	72.665	69.236	60.482	56.487	55.770	74.137	64.740	65.924	59.994	66.276	68.655	50.415	67.457	57.955	65.644	69.527	64.754	57.146
	H	U	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000
	H	S	1.000	1.000	0.750	1.000	1.000	1.000	0.500	1.000	1.000	1.000	1.000	0.750	1.000	1.000	1.000	0.750	1.000	1.000	1.000	1.000	1.000	1.000	0.750	1.000	0.500	1.000	0.750	1.000	1.000	0.750	1.000	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.750	1.000	0.750	1.000	1.000	1.000	0.750
	H	R	72.199	31.416	31.913	44.479	21.369	106.779	36.218	141.127	136.675	-8.638	108.599	44.673	85.467	97.607	75.562	38.307	57.076	47.900	-5.651	100.596	88.522	57.705	-0.647	137.681	50.829	105.988	34.485	71.418	62.141	4.417	101.010	99.246	128.606	105.246	83.658	49.940	0.604	67.531	84.783	33.344	71.351	72.141	85.202	57.798	106.561	20.273	58.571	108.634	30.672	64.310
	H	W	1.000	1.000	0.667	0.750	0.800	0.833	0.714	0.750	0.778	0.800	0.818	0.750	0.769	0.786	0.800	0.750	0.765	0.778	0.789	0.800	0.810	0.818	0.783	0.792	0.760	0.769	0.741	0.750	0.759	0.733	0.742	0.719	0.727	0.735	0.743	0.750	0.757	0.763	0.769	0.775	0.780	0.786	0.791	0.773	0.778	0.761	0.766	0.771	0.776	0.760

M：A 为 AT，F 为 FEB，G 为 GAT，H 为 HGAT；K：D 为平均拦截距离，U 和 S 分别为敌方无人机、无人艇拦截率，R 为回合奖励，W 为任务胜率。

表 D.3: 第四章目标选择多弹药配置测试原始数据

Cfg	M	K	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49
1-2-6-4	P	D	48.321	44.788	45.396	45.738	50.295	48.739	54.229	49.745	54.309	54.140	52.439	50.725	59.997	41.312	44.497	44.024	50.558	42.829	56.695	46.225	48.096	49.013	51.182	51.996	40.203	54.313	44.822	48.000	41.894	55.247	56.364	54.713	45.695	48.308	53.043	50.355	47.286	45.534	47.265	47.035	48.136	48.543	54.168	49.947	50.871	47.758	47.247	46.400	45.941	52.916
	P	F	0.237	0.234	0.222	0.213	0.262	0.217	0.267	0.250	0.267	0.273	0.268	0.250	0.261	0.227	0.250	0.231	0.286	0.256	0.300	0.257	0.244	0.184	0.267	0.239	0.278	0.314	0.256	0.238	0.290	0.279	0.279	0.224	0.273	0.220	0.279	0.220	0.244	0.175	0.240	0.256	0.208	0.239	0.275	0.261	0.234	0.265	0.227	0.233	0.343	0.240
	P	R	1.253	-2.324	-3.274	1.935	6.432	4.468	11.442	12.630	14.981	17.442	0.567	9.259	18.824	3.575	-1.568	-2.727	14.023	-2.242	18.958	0.320	10.951	-5.246	12.433	10.189	-2.106	-0.816	0.052	-1.362	-1.983	14.210	16.584	3.691	10.305	10.146	14.339	-0.059	-2.822	-1.838	14.006	-1.121	0.649	10.372	5.096	15.166	5.532	2.597	0.163	-0.482	11.751	15.387
	P	U	0.750	1.000	0.875	0.875	1.000	0.875	1.000	0.875	1.000	1.000	0.875	1.000	1.000	0.875	1.000	0.750	1.000	1.000	1.000	0.875	1.000	0.750	1.000	0.875	0.875	1.000	0.875	0.750	0.875	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	0.500	1.000	0.875	0.875	1.000	0.875	1.000	0.875	0.750	0.875	0.875	1.000	1.000
	P	S	0.750	0.750	0.750	0.750	0.750	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.750	0.750	0.750	1.000	0.500	1.000	0.500	0.750	0.750	1.000	1.000	0.750	0.750	0.750	1.000	0.500	1.000	1.000	1.000	1.000	0.750	1.000	0.750	0.750	0.750	1.000	0.750	0.750	0.750	1.000	1.000	1.000	0.750	0.750	0.750	1.000	1.000
	P	X	0.750	1.000	0.875	0.625	0.625	0.500	0.750	0.750	0.750	0.625	0.750	0.875	0.750	0.750	0.875	0.875	0.750	0.750	0.625	0.750	0.875	1.000	0.625	0.875	0.750	0.875	0.750	0.875	0.750	0.875	0.625	0.750	1.000	0.875	0.750	0.750	0.875	0.750	0.750	0.875	0.875	0.875	0.625	0.750	0.625	0.625	0.625	0.875	0.750	0.625
	P	Y	0.500	1.000	1.000	0.500	0.500	0.500	0.750	0.750	0.500	0.500	1.000	1.000	0.000	0.500	1.000	0.750	0.750	1.000	0.250	0.500	0.500	1.000	1.000	0.750	1.000	1.000	0.750	0.750	1.000	0.500	0.500	0.500	0.750	0.750	0.750	1.000	1.000	0.500	0.750	0.750	0.500	0.750	0.500	0.500	0.500	0.750	0.750	0.750	0.750	0.750
	Q	D	50.528	53.048	46.837	54.068	44.241	49.884	54.833	47.894	52.000	45.831	56.780	53.543	48.681	49.470	58.010	49.797	57.911	46.049	55.939	53.192	51.741	41.680	44.525	49.645	45.333	54.245	54.454	48.795	51.217	51.597	44.878	60.328	53.039	45.739	48.375	46.366	46.118	53.400	39.121	49.682	48.344	51.586	40.508	58.715	48.230	39.804	55.697	50.140	55.561	42.947
	Q	F	0.275	0.316	0.256	0.308	0.300	0.297	0.308	0.333	0.308	0.324	0.324	0.308	0.308	0.279	0.286	0.300	0.353	0.300	0.300	0.267	0.357	0.222	0.333	0.286	0.273	0.375	0.308	0.273	0.324	0.293	0.263	0.286	0.324	0.375	0.306	0.278	0.314	0.364	0.231	0.387	0.333	0.300	0.320	0.343	0.323	0.350	0.324	0.324	0.375	0.268
	Q	R	0.968	13.302	12.216	11.712	10.645	-0.997	14.519	3.888	13.041	-0.691	16.216	11.562	15.863	12.452	18.732	13.617	15.112	13.512	15.181	17.016	3.315	-2.402	-5.839	7.169	-2.885	11.661	12.594	15.883	14.394	11.794	-1.750	13.260	13.767	16.006	1.644	-0.169	-1.110	16.938	-6.078	10.225	0.812	13.402	-5.934	12.772	0.915	0.820	18.192	13.190	19.492	-2.880
	Q	U	1.000	1.000	0.875	1.000	1.000	1.000	1.000	0.875	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	0.750	1.000	0.875	0.750	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	0.875	1.000	1.000	0.875	1.000	1.000	1.000	0.750	1.000	0.875	0.625	1.000	1.000	1.000	1.000
	Q	S	0.750	1.000	1.000	1.000	1.000	0.750	1.000	0.500	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.750	0.500	0.250	0.750	0.750	1.000	1.000	1.000	1.000	1.000	0.750	1.000	1.000	1.000	0.750	0.750	0.750	1.000	0.500	1.000	0.750	1.000	0.500	1.000	0.750	0.500	1.000	1.000	1.000	0.750
	Q	X	0.750	0.750	0.625	0.875	0.875	0.875	0.750	0.500	0.750	0.875	0.750	0.750	0.750	0.750	0.625	0.750	0.750	0.750	0.750	0.750	0.625	0.750	1.000	0.375	1.000	0.750	0.750	0.625	0.625	0.750	0.875	0.875	0.750	0.750	0.750	0.750	1.000	0.625	1.000	0.875	0.750	0.875	0.750	0.750	0.750	0.500	0.625	0.750	0.500	1.000
	Q	Y	0.750	0.750	0.750	0.750	0.750	0.750	0.750	0.750	0.750	0.750	0.500	1.000	0.500	0.750	0.250	0.750	0.500	0.750	0.500	0.250	0.500	0.750	1.000	0.500	0.750	1.000	0.500	0.500	0.750	1.000	0.750	0.500	0.500	0.500	0.750	0.750	0.500	0.500	1.000	0.750	0.750	0.500	1.000	0.750	0.750	0.500	0.250	0.750	0.500	1.000
	M	D	51.153	53.355	59.075	56.254	60.400	52.025	53.893	63.173	54.383	48.359	52.531	44.782	56.776	49.115	41.282	40.911	57.706	49.796	56.722	59.722	52.469	44.628	44.799	55.012	60.918	54.646	55.177	46.771	50.259	53.262	57.789	55.910	49.406	51.783	48.587	57.068	51.565	57.342	53.069	49.071	55.375	51.658	53.698	53.160	52.831	47.064	49.248	44.783	52.270	54.586
	M	F	0.400	0.353	0.400	0.387	0.364	0.344	0.364	0.353	0.400	0.367	0.353	0.314	0.324	0.387	0.346	0.333	0.300	0.414	0.333	0.324	0.333	0.370	0.321	0.353	0.353	0.400	0.324	0.355	0.282	0.316	0.316	0.333	0.300	0.316	0.429	0.462	0.343	0.333	0.375	0.375	0.333	0.343	0.353	0.324	0.300	0.364	0.275	0.296	0.333	0.316
	M	R	13.866	12.935	16.912	14.169	14.603	15.633	15.560	17.442	14.457	14.847	16.373	-4.298	12.909	13.631	-2.369	-4.874	14.902	10.917	15.278	10.911	13.512	0.697	0.373	16.141	15.763	15.727	16.669	-1.280	-0.680	13.990	15.262	13.147	11.410	13.203	11.924	17.690	15.545	13.366	15.684	13.466	13.935	14.140	12.612	13.109	10.844	13.905	16.180	5.013	17.035	11.881
	M	U	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	0.875	1.000	1.000	0.875	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	0.875	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	0.750	1.000	1.000
	M	S	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.750	1.000	1.000	0.500	0.250	1.000	1.000	1.000	1.000	1.000	0.750	0.750	1.000	1.000	1.000	1.000	0.750	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.500	1.000	1.000
	M	X	0.750	0.875	0.625	0.875	0.750	0.625	0.875	0.625	0.750	0.500	0.750	1.000	0.750	0.875	0.875	0.875	0.750	0.875	0.625	0.750	0.625	0.750	0.625	0.750	0.750	0.625	0.625	0.875	0.875	0.750	0.625	0.750	0.875	0.875	0.875	0.625	0.625	0.750	0.625	0.875	0.875	0.750	0.875	0.750	0.875	0.750	0.625	0.500	0.625	0.750
	M	Y	0.750	0.500	0.750	0.500	0.750	0.500	0.250	0.500	0.500	0.750	0.250	1.000	0.750	0.500	0.500	1.000	0.500	0.750	0.750	1.000	0.750	0.500	0.750	0.500	0.250	0.750	0.750	1.000	1.000	0.750	0.750	0.750	0.500	0.500	0.750	0.500	0.750	0.750	0.500	0.750	0.500	0.500	0.500	0.750	0.750	0.500	0.500	0.250	0.500	0.750
2-1-6-4	P	D	45.563	47.191	41.873	49.043	41.084	53.401	53.195	49.837	44.876	49.800	47.048	44.302	52.964	47.242	44.974	47.243	52.722	48.464	55.478	49.459	51.830	58.415	51.152	49.437	47.511	51.306	54.487	56.913	51.172	48.399	52.526	54.443	59.154	46.242	49.643	46.078	52.034	46.171	54.499	46.907	51.989	53.929	51.722	49.499	51.990	55.601	52.639	51.778	51.848	47.909
	P	F	0.217	0.240	0.216	0.244	0.244	0.255	0.218	0.224	0.200	0.234	0.217	0.204	0.235	0.213	0.211	0.204	0.244	0.224	0.279	0.240	0.229	0.279	0.245	0.239	0.200	0.224	0.226	0.273	0.212	0.196	0.293	0.229	0.261	0.237	0.256	0.224	0.234	0.200	0.267	0.208	0.212	0.255	0.244	0.204	0.240	0.261	0.261	0.262	0.204	0.273
	P	R	1.450	10.136	1.873	-6.328	-2.979	13.776	11.173	-2.001	-0.097	-1.570	-0.148	10.095	15.552	-3.187	-1.972	11.416	3.353	3.969	15.909	16.771	11.672	15.802	11.214	1.337	2.685	14.468	18.778	14.020	15.484	-0.984	12.987	5.755	17.436	1.341	-1.213	-3.334	2.554	-0.061	17.687	-2.859	5.104	16.382	-1.702	-0.508	15.720	17.126	15.588	0.516	2.689	12.947
	P	U	0.875	1.000	0.750	0.875	1.000	1.000	1.000	1.000	0.875	1.000	0.875	1.000	1.000	0.875	0.750	1.000	0.875	0.875	1.000	1.000	0.875	1.000	1.000	1.000	0.750	0.875	1.000	1.000	0.875	0.875	1.000	0.875	1.000	0.750	0.750	0.875	0.875	0.875	1.000	0.875	0.875	1.000	1.000	0.875	1.000	1.000	1.000	1.000	0.875	1.000
	P	S	0.750	1.000	0.500	0.750	0.500	1.000	1.000	0.750	0.500	0.750	0.750	0.750	1.000	0.750	0.500	0.750	0.750	1.000	1.000	1.000	1.000	1.000	1.000	0.750	0.750	1.000	1.000	1.000	1.000	0.500	1.000	1.000	1.000	0.750	1.000	1.000	1.000	0.750	1.000	0.750	1.000	1.000	0.750	0.750	1.000	1.000	1.000	0.750	0.750	1.000
	P	X	0.750	0.875	0.500	1.000	0.875	0.875	0.875	0.875	0.750	1.000	0.875	0.750	0.750	1.000	0.750	0.875	0.625	0.750	0.750	0.625	0.750	0.875	0.750	0.750	0.625	0.625	0.625	0.750	0.625	0.875	0.750	0.625	0.625	0.750	0.750	1.000	0.875	0.750	0.625	1.000	0.875	0.750	1.000	0.750	0.875	0.750	0.750	0.750	0.750	0.625
	P	Y	0.750	1.000	0.750	1.000	1.000	0.500	0.500	1.000	0.750	1.000	0.750	0.750	0.500	1.000	0.750	0.750	0.750	0.500	0.500	0.500	0.750	0.500	0.750	1.000	0.750	0.750	0.250	0.750	0.500	0.750	0.750	0.500	0.500	0.500	0.750	1.000	0.500	0.750	0.500	1.000	0.000	0.500	1.000	0.750	0.250	0.250	0.500	1.000	0.500	1.000
	Q	D	40.557	50.796	50.063	58.345	45.012	42.350	48.812	55.657	54.180	44.712	49.173	49.643	56.085	45.515	51.164	44.728	49.420	50.791	44.411	38.709	43.976	54.190	55.177	49.991	47.088	57.557	48.784	58.116	50.068	45.157	40.741	52.953	55.946	52.968	57.027	49.053	43.518	50.453	50.946	41.996	54.147	39.253	50.684	46.274	46.045	47.222	55.363	55.496	53.130	57.178
	Q	F	0.256	0.293	0.324	0.273	0.286	0.289	0.387	0.333	0.343	0.224	0.333	0.308	0.324	0.344	0.333	0.387	0.357	0.293	0.303	0.286	0.314	0.308	0.300	0.275	0.345	0.293	0.312	0.300	0.333	0.323	0.333	0.267	0.316	0.375	0.387	0.333	0.222	0.333	0.333	0.256	0.316	0.278	0.263	0.323	0.297	0.286	0.300	0.364	0.370	0.353
	Q	R	11.841	13.798	12.179	16.965	-6.020	-2.306	14.415	14.479	17.910	2.559	12.226	11.281	15.154	3.909	1.317	12.678	-3.928	12.141	-4.851	-9.791	-1.513	17.399	16.173	0.098	-3.675	15.085	3.225	17.788	10.863	-2.208	-1.958	13.309	16.674	11.041	16.973	1.446	-5.786	12.031	17.587	-1.319	14.191	-1.206	0.127	-0.718	-0.316	2.575	11.886	13.712	2.797	15.498
	Q	U	1.000	1.000	1.000	1.000	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.625	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.750	1.000	1.000	0.875	1.000
	Q	S	0.750	1.000	1.000	1.000	0.500	0.750	1.000	1.000	1.000	0.750	1.000	1.000	1.000	0.750	0.750	1.000	0.500	1.000	0.500	0.250	0.750	1.000	1.000	0.750	0.500	1.000	0.750	1.000	1.000	0.500	0.750	1.000	1.000	1.000	1.000	0.750	0.500	1.000	1.000	0.500	1.000	0.500	0.500	0.500	0.750	1.000	1.000	1.000	0.750	1.000
	Q	X	0.750	0.875	0.875	0.625	0.875	1.000	0.625	0.750	0.500	0.750	0.625	0.875	0.750	0.750	0.875	0.875	0.875	0.875	1.000	0.875	0.750	0.750	0.750	0.875	0.875	0.625	0.750	0.750	0.875	1.000	1.000	0.875	0.750	0.750	0.625	0.750	0.875	0.750	0.625	0.750	0.875	0.875	0.875	0.750	1.000	0.750	0.875	0.875	0.625	0.750
	Q	Y	0.750	0.500	0.500	0.500	1.000	1.000	0.750	0.500	0.750	0.750	1.000	0.750	0.500	0.500	0.750	0.750	1.000	0.750	1.000	1.000	1.000	0.250	0.250	0.750	1.000	0.500	0.500	0.250	1.000	1.000	1.000	0.500	0.500	1.000	0.500	0.750	1.000	0.750	0.500	1.000	0.500	1.000	0.750	1.000	1.000	0.750	0.750	0.500	0.750	0.500
	M	D	52.231	58.819	57.125	50.547	49.989	56.957	42.898	41.821	49.769	55.346	53.235	52.099	54.315	51.208	49.545	53.275	52.066	49.648	49.433	52.894	60.326	58.147	47.707	47.350	50.893	52.760	56.163	47.036	53.489	50.439	53.882	58.192	60.675	45.207	61.054	47.753	58.686	53.555	54.534	53.734	52.165	47.119	53.917	56.302	52.188	50.590	53.344	53.952	54.976	52.305
	M	F	0.353	0.429	0.353	0.306	0.364	0.364	0.370	0.357	0.400	0.333	0.364	0.262	0.294	0.364	0.244	0.343	0.333	0.367	0.333	0.375	0.387	0.308	0.308	0.290	0.316	0.343	0.387	0.344	0.267	0.275	0.343	0.364	0.343	0.308	0.375	0.303	0.333	0.343	0.316	0.343	0.333	0.275	0.357	0.364	0.267	0.333	0.316	0.353	0.316	0.316
	M	R	15.961	16.509	14.215	11.650	19.337	11.584	-1.026	-3.920	10.742	17.216	16.797	3.169	4.923	15.795	12.640	11.847	1.217	3.363	16.240	14.796	15.464	15.135	14.200	0.420	15.352	14.456	12.892	-2.427	15.261	9.655	15.185	16.207	18.881	11.980	18.925	1.085	13.471	15.310	17.571	15.611	17.026	1.996	4.335	17.254	15.018	16.329	17.086	9.898	13.189	14.225
	M	U	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	0.875	1.000	0.875	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	M	S	1.000	1.000	1.000	0.750	1.000	1.000	0.500	0.500	1.000	1.000	1.000	1.000	0.750	1.000	1.000	1.000	1.000	0.750	1.000	1.000	1.000	1.000	1.000	0.500	1.000	1.000	1.000	0.750	1.000	0.750	1.000	1.000	1.000	1.000	1.000	0.750	1.000	1.000	1.000	1.000	1.000	0.750	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	M	X	0.625	0.750	0.750	0.875	0.625	0.750	0.875	0.750	0.875	0.750	0.750	0.875	0.625	0.625	0.750	0.875	0.875	0.750	0.750	0.750	0.750	0.750	0.625	0.625	0.750	0.875	0.750	1.000	0.750	0.875	0.750	0.750	0.625	0.750	0.625	0.750	0.875	0.750	0.500	0.875	0.750	0.875	0.625	0.625	0.625	0.875	0.625	0.875	0.875	0.875
	M	Y	0.750	0.500	0.750	0.750	0.250	1.000	1.000	1.000	1.000	0.250	0.250	0.500	0.500	0.750	0.750	0.750	0.500	0.500	0.500	0.500	0.500	0.500	0.750	0.750	0.500	0.500	0.750	0.750	0.500	0.750	0.500	0.500	0.250	1.000	0.500	0.750	0.500	0.500	0.500	0.250	0.250	0.500	0.500	0.500	0.750	0.250	0.500	1.000	0.500	0.500
2-2-3-4	P	D	42.265	42.815	47.317	48.859	40.087	44.964	41.537	42.587	49.125	46.719	44.657	45.889	44.207	50.507	50.926	49.100	49.977	59.939	49.813	48.303	56.594	39.384	50.836	44.124	46.059	53.527	37.922	37.852	33.412	51.632	48.291	48.590	42.974	37.526	46.204	46.502	39.834	41.413	44.873	43.161	49.641	39.385	43.464	46.621	47.152	50.892	46.111	45.049	41.700	33.361
	P	F	0.250	0.250	0.256	0.270	0.195	0.174	0.227	0.205	0.208	0.255	0.265	0.257	0.222	0.244	0.239	0.229	0.229	0.256	0.268	0.293	0.229	0.219	0.286	0.213	0.244	0.273	0.186	0.226	0.161	0.279	0.190	0.220	0.184	0.200	0.262	0.231	0.205	0.184	0.244	0.200	0.239	0.238	0.244	0.233	0.200	0.286	0.229	0.244	0.275	0.195
	P	R	1.715	9.824	11.841	1.789	-2.195	1.009	-3.501	-2.018	3.766	14.065	-0.895	0.893	-0.015	2.514	11.221	-0.893	12.981	16.632	1.921	15.169	5.894	0.528	13.293	11.879	12.497	14.204	-1.417	-3.290	-10.244	13.293	0.157	-0.836	-6.168	-4.174	10.790	0.963	-3.096	-6.148	2.528	-0.288	11.859	0.948	-1.935	-1.113	-0.404	15.151	1.201	9.955	-2.217	-4.031
	P	U	0.875	1.000	0.875	0.875	0.625	0.625	0.875	0.625	0.750	1.000	0.875	0.625	0.625	0.875	0.875	0.625	0.875	0.875	0.875	1.000	0.875	0.625	1.000	0.750	0.875	1.000	0.750	0.625	0.500	1.000	0.500	0.750	0.500	0.625	0.875	0.750	0.625	0.625	0.875	0.625	0.875	0.875	0.875	0.875	0.500	1.000	0.625	0.875	0.875	0.875
	P	S	0.750	1.000	1.000	0.750	0.750	0.750	0.750	1.000	1.000	1.000	0.500	1.000	0.750	1.000	1.000	0.750	1.000	1.000	1.000	1.000	1.000	0.500	1.000	1.000	1.000	1.000	0.500	0.500	0.250	1.000	1.000	0.750	0.750	0.500	1.000	0.750	0.750	0.500	0.750	0.500	1.000	0.750	0.750	0.750	1.000	1.000	0.750	1.000	1.000	0.250
	P	X	0.750	0.875	0.750	0.625	0.625	0.500	1.000	0.750	0.625	0.625	0.625	0.625	0.500	0.625	0.750	0.625	0.625	0.500	0.625	0.625	0.625	0.375	0.750	0.625	0.625	0.625	0.625	0.625	0.875	0.625	0.625	0.750	0.875	0.750	0.750	0.625	0.750	0.750	0.375	0.500	0.625	0.500	0.875	0.750	0.750	0.750	0.750	0.750	0.875	0.500
	P	Y	0.750	1.000	0.750	0.750	1.000	0.750	1.000	0.750	0.500	0.750	0.750	0.750	0.750	0.500	0.750	0.750	0.750	0.500	1.000	0.500	0.500	0.750	0.750	0.500	1.000	0.750	0.750	0.750	1.000	1.000	0.750	0.750	1.000	0.750	0.750	0.750	0.750	1.000	1.000	0.500	0.750	1.000	0.750	1.000	0.750	0.500	0.250	0.750	1.000	1.000
	Q	D	41.816	43.741	49.498	52.862	46.143	47.404	45.728	51.434	48.664	34.261	42.023	46.464	42.836	44.716	47.347	47.565	45.105	47.928	48.581	41.600	49.088	49.041	44.809	42.496	47.831	44.517	40.878	44.445	42.246	38.376	40.768	48.622	45.806	49.775	51.552	45.596	48.727	45.508	45.812	47.599	48.907	33.539	51.083	45.104	39.259	43.700	43.212	47.729	45.207	46.710
	Q	F	0.265	0.231	0.243	0.279	0.257	0.333	0.344	0.239	0.250	0.250	0.195	0.262	0.233	0.308	0.243	0.244	0.214	0.333	0.275	0.237	0.303	0.286	0.262	0.303	0.243	0.242	0.250	0.256	0.235	0.167	0.281	0.297	0.238	0.227	0.278	0.216	0.211	0.225	0.250	0.233	0.265	0.240	0.244	0.278	0.233	0.300	0.250	0.265	0.276	0.303
	Q	R	-2.216	0.129	1.986	15.729	2.626	13.854	10.172	-1.590	2.919	-6.454	-2.428	0.272	0.717	2.968	0.458	1.764	-3.046	4.005	10.312	-1.890	0.137	11.527	2.456	-0.670	-0.353	-1.774	-3.460	0.851	-1.034	-5.964	-1.290	0.321	4.316	2.359	2.242	0.853	-1.097	-0.207	-0.998	-1.628	-0.297	-2.601	14.400	-0.221	-4.123	-2.851	-0.826	0.063	-2.989	-1.990
	Q	U	0.750	0.750	0.625	1.000	0.750	1.000	0.875	0.875	0.750	0.500	0.625	1.000	0.750	0.625	0.625	0.875	0.625	0.875	0.875	0.750	0.875	1.000	1.000	0.875	0.750	0.625	0.625	0.875	0.625	0.500	0.875	1.000	0.750	0.750	0.750	0.625	0.625	0.750	0.750	0.875	0.750	0.625	0.875	0.750	0.750	0.750	0.750	0.750	0.625	0.875
	Q	S	0.750	0.750	1.000	1.000	0.750	1.000	1.000	1.000	0.750	0.500	0.750	0.750	1.000	0.750	1.000	1.000	1.000	0.750	1.000	0.750	0.750	1.000	0.750	0.750	0.750	0.750	0.500	0.750	0.750	0.500	0.500	0.750	1.000	1.000	1.000	0.750	0.750	0.750	0.750	0.750	0.750	0.250	1.000	1.000	0.250	0.750	0.750	0.750	0.750	0.750
	Q	X	0.750	0.625	0.750	0.500	0.625	0.625	0.625	0.875	0.500	0.625	0.750	0.625	0.625	0.500	0.625	0.750	0.750	0.625	0.750	0.750	0.750	0.625	0.625	0.750	0.625	0.625	0.625	0.625	0.500	0.750	0.875	0.750	0.500	0.625	0.625	0.500	0.625	0.625	0.625	0.875	0.750	0.500	0.625	0.875	0.625	0.875	0.500	0.625	0.875	0.750
	Q	Y	1.000	0.750	0.500	0.750	0.500	0.750	1.000	0.750	0.500	1.000	1.000	1.000	1.000	0.250	0.750	0.750	1.000	0.750	0.750	1.000	0.750	0.750	0.750	0.750	0.750	1.000	0.750	0.750	1.000	1.000	0.750	0.750	0.500	1.000	0.750	0.750	0.750	1.000	0.750	1.000	0.500	1.000	0.750	0.750	1.000	0.750	0.750	1.000	0.500	1.000
	M	D	49.111	50.532	43.922	48.288	41.475	46.821	40.896	51.385	44.522	47.191	47.451	39.998	47.012	44.064	45.508	46.523	49.468	46.575	54.324	47.917	47.202	38.611	46.822	39.553	41.726	45.885	49.678	43.577	46.898	45.135	43.692	45.024	42.053	45.359	47.419	49.092	46.968	43.402	44.282	41.673	53.219	44.161	47.883	43.298	44.184	47.956	45.671	41.880	47.823	42.393
	M	F	0.275	0.273	0.343	0.314	0.276	0.286	0.250	0.314	0.257	0.306	0.310	0.222	0.263	0.281	0.312	0.312	0.306	0.321	0.300	0.256	0.294	0.333	0.303	0.312	0.333	0.294	0.324	0.314	0.250	0.212	0.355	0.296	0.280	0.289	0.258	0.333	0.294	0.194	0.333	0.226	0.387	0.292	0.265	0.231	0.294	0.256	0.333	0.241	0.393	0.250
	M	R	3.079	1.574	13.470	5.517	0.360	-0.809	-4.087	9.766	1.650	-0.792	3.241	-2.187	1.601	0.776	-0.720	0.735	0.496	-0.468	12.951	2.008	0.148	-5.315	-1.875	1.663	-1.825	0.413	13.001	0.717	0.365	-2.844	-1.223	2.679	2.626	0.456	0.881	10.599	3.437	-5.684	0.115	-3.095	16.516	-1.925	0.117	-0.617	-0.963	-0.522	-0.697	-3.909	11.183	-1.999
	M	U	1.000	0.625	1.000	1.000	0.625	0.875	0.625	0.875	0.750	0.875	0.875	0.625	0.750	0.750	0.750	0.875	1.000	0.750	1.000	0.750	0.750	0.750	0.750	0.875	0.750	0.875	0.875	1.000	0.625	0.625	1.000	0.625	0.500	0.875	0.625	1.000	0.750	0.500	1.000	0.625	1.000	0.625	0.750	0.750	1.000	0.875	0.500	0.500	0.875	0.625
	M	S	0.750	1.000	1.000	0.750	0.750	0.750	0.750	1.000	0.750	1.000	0.500	0.750	1.000	0.750	1.000	0.750	0.750	0.750	1.000	1.000	1.000	0.750	1.000	0.750	0.750	0.750	1.000	0.750	0.750	0.500	0.750	0.750	0.750	1.000	0.750	1.000	1.000	0.750	0.750	0.500	1.000	0.500	0.750	0.750	0.500	0.750	1.000	0.750	1.000	0.750
	M	X	0.500	0.750	0.500	0.500	0.500	0.500	0.750	0.750	0.500	0.750	0.500	0.625	0.625	0.625	0.625	0.750	0.875	0.750	0.750	0.625	0.750	0.875	0.875	0.625	0.750	0.750	0.625	0.625	0.500	0.875	0.750	0.500	0.625	0.625	0.500	0.750	0.625	0.625	0.625	0.375	0.625	0.750	0.625	0.625	0.750	0.750	0.750	0.625	0.625	0.750
	M	Y	0.750	0.500	0.750	0.500	0.750	1.000	1.000	0.750	1.000	1.000	0.500	1.000	1.000	0.750	1.000	0.500	0.500	0.500	0.750	0.750	0.750	1.000	1.000	0.750	1.000	0.750	0.750	1.000	0.750	0.500	1.000	0.500	0.250	1.000	0.750	1.000	0.500	1.000	0.750	1.000	0.500	0.500	0.750	0.750	1.000	0.750	0.500	0.750	0.750	0.750
2-2-6-2	P	D	41.716	51.445	49.826	57.686	46.544	49.511	54.568	47.861	48.285	54.339	55.221	60.171	46.911	54.422	56.285	56.507	52.291	49.308	46.299	53.444	53.057	49.769	40.772	45.972	56.704	50.452	50.562	50.688	50.384	48.842	52.983	45.978	52.028	54.394	53.736	49.825	51.728	50.154	39.518	52.816	51.464	50.972	47.553	47.484	50.834	55.698	54.934	48.813	50.000	48.741
	P	F	0.162	0.250	0.273	0.235	0.234	0.231	0.255	0.245	0.216	0.240	0.224	0.255	0.200	0.239	0.286	0.255	0.235	0.279	0.222	0.229	0.255	0.261	0.233	0.235	0.226	0.255	0.235	0.212	0.255	0.212	0.267	0.204	0.275	0.282	0.250	0.234	0.255	0.256	0.196	0.262	0.231	0.224	0.229	0.216	0.222	0.250	0.245	0.244	0.227	0.244
	P	R	-5.906	8.783	17.419	13.759	7.476	5.258	13.398	11.798	12.315	15.751	1.268	12.559	0.916	14.786	17.261	17.864	15.134	18.377	3.859	0.631	13.665	17.086	-0.073	13.031	14.231	11.593	10.830	3.128	14.262	0.877	14.796	10.774	4.159	7.368	11.595	4.175	11.413	3.275	0.634	1.273	1.475	11.981	-0.737	-1.171	10.200	12.999	12.372	0.504	2.280	9.625
	P	U	0.625	1.000	1.000	1.000	1.000	0.750	1.000	1.000	0.875	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	0.875	0.875	0.750	1.000	0.625	1.000	1.000	1.000	1.000	1.000	0.750	1.000
	P	S	0.250	1.000	1.000	1.000	0.750	0.750	1.000	1.000	1.000	1.000	0.750	1.000	0.750	0.750	1.000	1.000	1.000	1.000	0.750	0.750	1.000	1.000	0.500	1.000	1.000	1.000	1.000	0.750	1.000	0.750	1.000	0.750	0.750	0.750	1.000	0.750	1.000	0.750	0.750	1.000	0.750	0.750	0.750	0.750	1.000	1.000	1.000	0.750	1.000	0.750
	P	X	0.750	1.000	0.500	0.750	1.000	0.625	0.875	0.875	0.750	0.750	0.750	0.750	0.875	0.750	0.750	0.625	0.625	0.500	0.500	1.000	0.750	0.500	0.875	0.750	0.750	0.875	0.750	0.750	0.750	0.875	0.750	0.875	0.750	0.500	1.000	0.750	1.000	0.500	0.625	0.875	0.750	0.750	0.750	1.000	1.000	0.875	0.750	0.875	0.875	1.000
	P	Y	0.500	0.750	0.750	0.750	0.750	0.250	0.500	0.750	0.500	0.500	0.750	0.500	0.500	0.500	0.500	0.500	0.750	0.500	0.750	0.500	0.750	0.750	0.500	0.750	0.500	0.750	0.750	0.500	0.500	0.500	0.500	0.750	0.500	0.500	0.500	0.500	0.500	1.000	0.750	0.500	0.500	0.500	0.500	1.000	0.750	0.500	0.750	0.500	0.500	0.500
	Q	D	50.151	46.863	53.157	53.449	51.130	50.600	53.066	54.169	54.705	56.803	58.534	45.347	51.126	54.187	48.178	45.748	55.092	56.087	53.175	53.720	48.734	45.947	51.485	46.324	49.960	54.062	52.993	44.848	50.705	50.553	54.748	39.486	48.012	41.771	55.048	52.908	52.207	49.576	48.337	42.068	51.181	51.236	37.338	55.798	38.498	54.147	55.815	52.292	40.790	44.881
	Q	F	0.300	0.275	0.308	0.324	0.250	0.300	0.267	0.324	0.353	0.324	0.293	0.333	0.316	0.300	0.333	0.345	0.273	0.286	0.324	0.316	0.268	0.273	0.300	0.270	0.279	0.308	0.344	0.279	0.303	0.344	0.293	0.333	0.300	0.250	0.293	0.286	0.343	0.263	0.293	0.296	0.324	0.289	0.200	0.316	0.229	0.308	0.267	0.282	0.250	0.250
	Q	R	13.959	-1.528	14.120	18.190	13.558	13.238	14.590	12.150	14.955	14.237	14.318	-0.827	12.558	17.491	2.331	0.762	17.028	17.032	14.258	16.174	10.331	11.109	14.811	1.257	10.154	11.651	1.280	13.027	1.959	6.344	17.460	-3.764	13.508	-2.019	13.999	15.232	10.574	1.079	17.140	-3.056	14.661	9.626	2.173	13.972	-3.417	14.309	16.680	4.719	-3.012	-0.118
	Q	U	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	0.875	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	0.875	1.000	0.750	1.000	1.000	0.625	1.000	0.750	1.000	1.000	1.000	0.750	0.750
	Q	S	1.000	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.750	1.000	1.000	0.750	0.750	1.000	1.000	1.000	1.000	0.750	1.000	1.000	0.750	1.000	1.000	0.750	1.000	0.750	0.750	1.000	0.500	1.000	0.500	1.000	1.000	1.000	0.750	1.000	0.500	1.000	0.750	0.250	1.000	0.500	1.000	1.000	0.750	1.000	0.750
	Q	X	0.750	1.000	0.875	0.625	0.750	0.750	0.875	0.875	0.750	0.750	1.000	0.875	0.875	0.625	0.750	0.750	0.625	0.750	0.750	0.750	1.000	0.750	0.750	0.875	1.000	0.875	1.000	0.875	0.750	0.625	0.625	0.625	0.750	0.875	0.750	0.750	0.875	0.750	0.625	0.750	0.750	0.875	0.375	0.875	0.625	0.750	0.750	0.750	0.875	0.750
	Q	Y	0.750	0.750	0.500	0.500	0.500	0.750	0.500	0.750	0.750	0.750	0.500	0.750	0.750	0.500	0.500	0.750	0.750	0.250	0.750	0.500	0.500	1.000	0.500	0.500	0.750	0.750	0.500	0.750	0.250	0.500	0.500	1.000	0.750	0.750	0.500	0.500	0.750	0.500	0.500	0.750	0.750	0.750	0.500	0.500	0.750	0.750	0.250	0.250	1.000	0.750
	M	D	46.903	52.318	56.966	49.128	52.703	51.620	54.915	49.327	55.910	50.873	52.443	51.145	49.267	55.766	56.355	50.108	45.929	50.603	48.437	57.715	49.536	45.439	50.340	49.691	53.799	51.377	60.148	57.617	48.281	49.766	51.448	53.615	52.838	61.304	43.317	43.278	45.783	48.274	54.978	52.328	56.427	49.899	52.694	57.975	49.842	55.583	54.772	51.543	61.058	53.597
	M	F	0.297	0.364	0.293	0.294	0.400	0.344	0.324	0.324	0.308	0.462	0.282	0.297	0.220	0.364	0.300	0.333	0.256	0.314	0.290	0.316	0.286	0.324	0.316	0.306	0.364	0.308	0.333	0.343	0.333	0.344	0.279	0.286	0.343	0.324	0.212	0.268	0.364	0.273	0.308	0.333	0.300	0.244	0.312	0.333	0.343	0.300	0.286	0.324	0.353	0.343
	M	R	11.569	16.081	13.951	-1.137	2.122	2.776	12.119	15.447	15.954	10.871	1.603	10.876	3.322	12.717	17.458	12.279	-0.384	11.454	-0.690	15.608	0.707	-4.551	12.622	5.904	13.932	14.470	17.338	18.587	14.832	-0.474	14.274	13.849	13.035	16.900	-1.890	1.648	12.191	13.054	15.039	15.542	16.041	11.559	3.714	14.830	10.684	14.745	18.420	14.083	20.071	14.957
	M	U	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.625	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	M	S	0.750	1.000	1.000	0.500	0.750	0.750	1.000	1.000	1.000	1.000	0.750	0.750	0.750	1.000	1.000	1.000	0.500	0.750	0.500	1.000	0.750	0.750	1.000	0.750	1.000	1.000	1.000	1.000	1.000	0.750	1.000	1.000	1.000	1.000	0.500	0.750	1.000	1.000	1.000	1.000	1.000	0.500	0.500	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	M	X	0.875	0.750	0.750	0.875	0.750	0.750	0.625	0.625	0.750	0.875	0.875	1.000	0.625	0.875	0.500	0.875	0.875	0.875	0.625	0.750	0.750	1.000	0.875	0.625	0.750	0.750	0.500	0.625	0.750	1.000	0.750	0.750	0.875	0.625	1.000	0.875	0.875	0.750	0.750	0.750	0.750	0.750	0.625	0.750	0.875	0.750	0.750	0.750	0.625	0.750
	M	Y	0.500	0.250	0.750	0.500	0.500	0.500	0.750	0.750	0.500	0.750	0.750	0.500	0.500	0.750	0.750	0.500	0.500	0.750	0.750	0.500	1.000	1.000	0.750	0.500	0.750	0.500	0.500	0.250	0.500	0.750	0.750	0.750	0.500	0.500	0.250	0.500	0.750	0.750	0.500	0.500	0.500	0.750	0.500	0.500	0.750	0.500	0.250	0.750	0.250	0.500

M：P 为 PPO，Q 为 QMIX，M 为 QMIX-MAT；K：D 为平均拦截距离，F 为弹药利用效率，R 为回合奖励，U 和 S 分别为敌方无人机、无人艇拦截率，X 和 Y 分别为我方无人机、无人艇损失率。

表 D.4: 续表——第四章目标选择多弹药配置测试原始数据

Cfg	M	K	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49
2-2-6-4	P	D	50.795	56.509	45.837	48.313	55.925	58.264	58.251	43.108	56.084	50.439	49.849	50.595	50.869	44.183	52.075	58.275	56.663	47.659	46.207	55.003	41.526	58.091	52.658	56.781	59.388	43.136	52.099	48.486	45.109	48.124	49.631	49.745	54.997	52.134	45.906	57.860	46.274	48.937	54.682	51.036	51.724	42.278	55.697	56.776	50.144	53.437	55.174	47.775	43.309	46.064
	P	F	0.222	0.250	0.214	0.267	0.273	0.279	0.216	0.235	0.226	0.196	0.235	0.214	0.224	0.208	0.189	0.245	0.240	0.208	0.220	0.218	0.196	0.218	0.261	0.245	0.279	0.213	0.261	0.245	0.204	0.200	0.216	0.200	0.234	0.267	0.226	0.250	0.196	0.196	0.267	0.209	0.267	0.163	0.255	0.235	0.208	0.218	0.226	0.212	0.209	0.220
	P	R	1.945	11.158	10.141	15.194	13.222	10.664	-1.065	1.022	11.878	-0.912	10.855	10.419	0.808	-4.590	-1.126	18.699	16.514	0.033	2.062	16.841	-4.277	13.080	11.017	15.909	18.727	-1.148	10.897	9.127	-1.459	0.017	-1.552	-0.971	0.392	14.388	11.677	17.739	2.237	9.138	15.290	5.488	11.872	-4.450	13.868	16.579	1.858	12.568	14.241	0.213	-2.381	10.210
	P	U	0.875	1.000	1.000	1.000	1.000	1.000	0.875	0.750	1.000	0.875	1.000	1.000	0.875	0.750	0.875	1.000	1.000	0.750	0.875	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	0.750	0.875	0.875	0.875	1.000	1.000	1.000	0.750	1.000	1.000	0.750	1.000	0.750	1.000	1.000	0.875	1.000	1.000	1.000	0.875	1.000
	P	S	0.750	1.000	1.000	1.000	1.000	1.000	1.000	0.500	1.000	0.500	1.000	1.000	1.000	1.000	0.750	1.000	1.000	1.000	1.000	1.000	0.500	1.000	1.000	1.000	1.000	0.750	1.000	1.000	0.750	0.750	1.000	0.750	1.000	1.000	1.000	1.000	1.000	0.750	1.000	0.750	1.000	0.500	1.000	1.000	0.750	1.000	1.000	0.750	0.500	0.750
	P	X	0.750	0.875	1.000	0.750	0.750	1.000	0.875	0.500	0.750	0.875	0.875	0.750	0.750	1.000	1.000	0.625	0.750	0.875	0.875	0.750	1.000	0.875	0.875	0.750	0.625	0.750	0.875	0.875	0.875	0.750	0.875	0.875	0.750	0.750	0.750	0.625	0.750	0.875	0.750	0.625	0.875	0.875	0.750	0.750	0.750	0.750	0.875	0.875	0.750	0.875
	P	Y	0.750	0.750	0.750	0.500	0.750	0.750	0.750	0.750	0.750	0.500	0.750	1.000	1.000	1.000	0.750	0.500	0.250	0.500	0.750	0.250	1.000	0.500	0.750	0.500	0.000	0.750	0.750	1.000	1.000	0.500	0.750	0.500	1.000	0.500	0.750	0.500	0.500	1.000	0.500	0.250	0.750	0.500	0.750	0.250	0.500	0.750	0.500	1.000	1.000	0.750
	Q	D	51.294	48.497	51.824	49.247	55.536	52.399	51.246	50.165	52.815	49.028	41.100	51.761	54.531	57.158	55.418	52.927	53.349	54.486	42.448	51.729	51.814	44.960	51.764	48.886	54.035	52.556	41.050	49.877	54.191	51.305	52.732	50.781	51.592	50.676	48.059	48.351	49.193	48.828	51.929	49.986	51.541	53.290	58.577	50.872	50.241	49.731	51.853	54.481	48.043	52.488
	Q	F	0.275	0.275	0.312	0.308	0.273	0.263	0.316	0.268	0.286	0.316	0.235	0.279	0.316	0.333	0.333	0.279	0.308	0.286	0.286	0.279	0.293	0.289	0.306	0.308	0.250	0.333	0.385	0.263	0.282	0.303	0.273	0.227	0.293	0.235	0.279	0.282	0.286	0.200	0.286	0.261	0.273	0.279	0.245	0.286	0.290	0.268	0.244	0.293	0.273	0.286
	Q	R	6.581	0.202	5.648	8.798	14.668	-0.296	13.028	-1.537	13.349	15.163	2.598	11.895	19.434	17.564	15.705	17.889	16.220	13.138	-3.948	12.822	14.320	-0.546	3.207	12.259	9.772	17.661	0.078	-0.097	3.585	0.022	18.190	-0.021	12.776	14.776	14.535	0.770	13.180	10.792	14.533	11.222	11.971	13.125	15.592	12.468	-1.190	-2.330	2.347	16.583	11.501	-1.186
	Q	U	1.000	1.000	0.875	1.000	1.000	0.750	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	0.875	1.000	1.000	1.000	0.875	1.000	1.000	0.875	1.000	0.875	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	0.750	1.000	0.875	1.000	1.000	1.000
	Q	S	0.750	0.750	0.750	1.000	1.000	1.000	1.000	0.750	1.000	1.000	0.250	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.250	1.000	1.000	0.750	1.000	1.000	1.000	1.000	0.750	0.500	0.750	0.750	1.000	0.750	1.000	1.000	1.000	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.750	0.750	0.750	1.000	1.000	1.000
	Q	X	0.625	0.750	0.750	0.875	0.875	0.875	0.750	0.875	0.875	0.750	0.500	0.875	0.500	0.500	0.750	0.625	0.625	0.875	0.750	0.875	0.750	0.875	0.750	0.750	1.000	0.625	0.625	0.875	0.750	0.750	0.750	0.750	0.875	0.750	0.750	0.750	0.875	0.875	0.625	0.875	0.875	0.750	0.750	0.875	0.750	0.875	0.750	0.625	0.750	1.000
	Q	Y	0.250	1.000	0.250	1.000	0.500	0.500	0.750	1.000	0.500	0.500	0.750	0.750	0.250	0.500	0.500	0.250	0.500	0.500	1.000	0.500	0.750	1.000	0.750	0.750	0.750	0.500	1.000	0.750	0.500	0.750	0.250	0.750	0.750	0.500	0.500	1.000	0.500	0.750	0.500	0.750	0.750	0.750	0.500	0.500	0.750	1.000	0.500	0.500	0.750	1.000
	M	D	52.072	57.404	47.665	56.076	61.975	53.591	49.336	48.708	53.072	49.314	51.848	53.817	50.152	55.242	54.926	57.958	52.497	49.048	49.261	53.753	55.271	56.536	61.432	61.167	51.436	56.395	47.332	44.746	47.999	51.393	50.427	50.045	57.695	53.893	57.360	51.418	41.482	53.887	51.675	44.993	52.380	53.809	47.067	53.525	56.292	50.694	50.429	51.935	57.142	55.829
	M	F	0.324	0.300	0.345	0.324	0.286	0.316	0.286	0.308	0.316	0.308	0.324	0.255	0.297	0.333	0.267	0.286	0.316	0.316	0.263	0.293	0.300	0.364	0.300	0.343	0.314	0.286	0.268	0.333	0.343	0.353	0.343	0.387	0.333	0.308	0.235	0.293	0.333	0.308	0.300	0.391	0.364	0.273	0.303	0.333	0.308	0.333	0.300	0.324	0.324	0.267
	M	R	13.279	14.373	1.020	17.016	15.786	15.605	16.726	13.182	13.631	13.374	16.939	13.114	-2.064	18.755	15.728	18.438	14.336	13.150	0.695	10.486	18.087	14.238	18.671	16.744	0.614	13.568	2.479	1.534	13.286	11.816	12.847	14.274	20.342	15.720	11.835	13.441	-2.624	16.172	16.305	2.974	13.024	11.506	-0.147	14.768	15.006	11.612	14.252	17.788	16.556	15.466
	M	U	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.750	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	M	S	1.000	1.000	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.750	1.000	1.000	1.000	1.000	1.000	0.750	1.000	1.000	1.000	1.000	1.000	0.750	1.000	0.750	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.500	1.000	1.000	0.750	1.000	1.000	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	M	X	0.750	0.750	0.750	0.625	0.875	0.750	0.625	0.750	0.750	0.750	0.625	0.875	1.000	0.500	0.750	0.625	0.750	0.750	0.625	1.000	0.625	0.625	0.625	0.750	0.875	0.750	0.750	0.625	0.750	0.875	0.875	0.750	0.375	0.625	0.750	0.750	0.875	0.750	0.625	0.625	0.875	0.875	0.750	0.750	0.750	0.875	0.750	0.750	0.875	0.750
	M	Y	0.750	0.750	0.500	0.250	0.250	0.500	0.500	0.750	0.750	0.750	0.500	0.500	1.000	0.500	0.250	0.250	0.500	0.750	0.750	0.750	0.500	0.750	0.250	0.250	0.750	0.500	1.000	1.000	0.750	0.750	0.750	0.750	0.500	0.500	0.750	0.750	1.000	0.250	0.500	0.500	0.750	0.750	1.000	0.750	0.500	0.750	0.500	0.250	0.250	0.500
3-2-6-4	P	D	47.901	45.940	51.259	51.428	51.146	50.986	44.514	54.509	48.623	56.011	45.963	50.630	55.087	45.588	47.391	44.913	53.757	53.390	53.969	47.017	51.506	47.482	49.819	50.397	50.873	40.644	46.194	59.030	58.322	47.425	58.810	45.708	47.190	48.295	52.931	42.195	47.613	53.839	49.344	47.477	43.623	45.662	52.239	47.421	42.196	47.021	49.445	43.298	40.394	51.919
	P	F	0.196	0.182	0.192	0.214	0.224	0.222	0.204	0.240	0.176	0.208	0.250	0.207	0.211	0.175	0.184	0.200	0.222	0.212	0.240	0.207	0.214	0.196	0.245	0.196	0.238	0.189	0.196	0.255	0.214	0.209	0.218	0.175	0.196	0.179	0.218	0.179	0.218	0.226	0.200	0.190	0.200	0.192	0.250	0.244	0.216	0.196	0.204	0.176	0.190	0.226
	P	R	0.876	1.151	-1.871	15.800	0.058	17.689	-0.282	13.154	-0.444	4.406	9.737	13.992	13.261	2.361	-1.633	-2.459	13.592	10.776	18.284	8.548	14.006	-4.065	14.171	0.453	5.048	-4.001	1.424	11.300	11.535	0.995	17.556	1.122	0.173	-2.745	12.611	-2.828	13.461	12.332	3.417	2.614	-2.193	-2.153	11.920	0.084	-4.326	2.048	-0.067	-1.980	-2.406	15.986
	P	U	0.875	0.875	0.875	1.000	1.000	1.000	0.750	1.000	0.750	0.875	1.000	1.000	1.000	0.875	0.750	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.750	1.000	0.875	1.000	1.000	0.750	1.000	0.875	1.000	0.750	1.000	1.000	1.000	1.000	0.875	1.000	0.875	1.000	1.000	1.000	1.000	0.750	0.875	0.750	0.750	1.000
	P	S	0.750	0.750	0.750	1.000	0.750	1.000	1.000	1.000	0.750	1.000	1.000	1.000	1.000	0.750	0.750	0.750	1.000	0.750	1.000	1.000	1.000	0.500	1.000	0.750	1.000	0.500	1.000	1.000	1.000	0.750	1.000	0.750	0.750	1.000	1.000	0.500	1.000	1.000	0.750	0.750	0.750	0.500	1.000	0.750	0.750	0.750	0.750	0.750	0.500	1.000
	P	X	0.750	0.750	1.000	0.750	0.875	0.625	0.875	0.875	0.750	0.750	1.000	0.875	0.875	0.750	0.625	0.750	0.750	0.875	0.625	1.000	0.875	0.875	0.750	0.875	0.625	1.000	0.750	0.875	0.875	0.625	0.750	0.875	1.000	1.000	0.875	0.875	0.875	0.875	0.625	0.875	0.875	0.875	0.875	0.875	1.000	0.750	0.875	0.875	0.625	0.750
	P	Y	0.750	0.500	0.750	0.500	1.000	0.500	0.750	0.750	0.750	0.500	0.750	0.500	0.500	0.500	1.000	1.000	0.750	0.750	0.500	0.750	0.500	1.000	0.500	1.000	0.500	1.000	0.750	0.750	0.750	0.750	0.250	0.500	0.750	0.750	0.750	1.000	0.750	0.750	0.500	0.500	1.000	1.000	0.750	1.000	1.000	0.500	0.750	0.500	1.000	0.500
	Q	D	48.249	53.077	53.941	46.329	50.946	52.840	57.594	50.127	43.642	51.785	57.109	52.776	57.106	42.036	52.898	51.988	52.311	58.036	55.080	47.298	50.909	52.093	49.887	54.492	46.100	51.310	44.617	53.935	47.355	53.931	43.464	58.271	54.191	56.968	49.004	56.130	50.498	46.289	54.247	50.564	41.834	55.571	45.446	56.391	51.829	51.925	52.194	53.813	53.659	57.392
	Q	F	0.256	0.273	0.255	0.256	0.279	0.250	0.273	0.267	0.243	0.270	0.267	0.324	0.267	0.241	0.267	0.268	0.300	0.279	0.324	0.300	0.239	0.255	0.279	0.231	0.217	0.238	0.234	0.273	0.273	0.240	0.220	0.279	0.286	0.300	0.275	0.261	0.275	0.222	0.273	0.240	0.235	0.255	0.250	0.279	0.293	0.267	0.279	0.364	0.286	0.300
	Q	R	0.926	16.468	10.745	-4.975	8.623	13.127	19.046	14.110	-3.112	2.254	16.565	12.657	15.269	-0.589	13.424	-3.472	11.829	16.748	13.323	11.963	2.899	10.422	12.196	17.053	-0.450	0.284	9.233	12.388	8.404	14.643	-0.802	19.819	15.216	14.553	2.773	13.911	10.940	11.099	13.323	14.691	-2.504	14.723	1.448	15.028	13.550	15.943	18.256	15.958	18.466	14.231
	Q	U	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	0.625	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.750	1.000	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	Q	S	0.750	1.000	1.000	0.500	1.000	1.000	1.000	1.000	0.500	0.500	1.000	1.000	1.000	0.500	1.000	1.000	1.000	1.000	1.000	1.000	0.750	1.000	1.000	1.000	0.750	0.750	0.750	1.000	1.000	1.000	0.750	1.000	1.000	1.000	0.750	1.000	0.750	1.000	1.000	1.000	0.500	1.000	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	Q	X	0.875	0.750	0.875	1.000	1.000	0.875	0.750	0.750	1.000	0.750	0.750	0.875	0.750	0.625	0.750	1.000	0.875	0.625	0.875	0.875	0.875	0.875	0.875	0.750	1.000	0.750	1.000	0.875	1.000	0.625	0.875	0.625	0.750	0.625	0.625	0.875	0.875	0.875	0.750	0.625	0.625	0.875	0.625	0.750	0.750	0.750	0.625	0.750	0.625	0.750
	Q	Y	0.750	0.500	0.750	1.000	0.750	0.500	0.250	0.750	0.500	0.750	0.500	0.500	0.500	0.500	0.750	1.000	0.750	0.250	0.500	0.750	0.250	0.750	0.750	0.500	0.500	0.750	0.750	0.750	0.750	0.750	0.750	0.250	0.750	0.750	0.750	0.250	0.750	0.750	0.750	0.500	0.750	0.250	0.750	0.500	0.750	0.500	0.250	0.500	0.250	0.500
	M	D	57.995	45.547	51.001	51.701	52.301	52.863	58.009	54.003	55.587	48.280	52.194	53.939	48.189	50.380	58.109	44.950	51.989	56.202	54.788	48.967	54.762	44.556	51.281	52.219	51.848	52.029	55.170	44.535	50.042	49.086	50.340	54.451	39.215	55.553	52.622	45.441	49.745	53.445	57.533	50.750	61.496	55.221	51.177	54.481	51.801	53.763	54.228	45.570	52.157	43.833
	M	F	0.343	0.250	0.364	0.261	0.293	0.333	0.300	0.308	0.308	0.279	0.316	0.300	0.273	0.316	0.267	0.333	0.286	0.308	0.245	0.333	0.306	0.273	0.282	0.286	0.308	0.293	0.286	0.367	0.316	0.353	0.224	0.245	0.290	0.324	0.235	0.375	0.308	0.343	0.250	0.308	0.400	0.293	0.324	0.316	0.385	0.343	0.293	0.345	0.286	0.286
	M	R	16.496	3.137	17.789	14.511	13.229	22.334	12.844	14.975	18.162	11.800	16.718	15.130	12.890	13.168	10.256	11.956	12.500	15.880	12.924	13.119	3.543	1.751	1.396	13.014	16.043	15.076	14.854	10.418	9.670	11.572	-1.521	14.612	-1.526	15.385	12.407	-1.381	11.518	17.653	15.187	14.168	22.679	18.940	13.555	16.745	3.537	13.307	18.278	-0.150	14.661	-2.191
	M	U	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	0.875	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	1.000	1.000	1.000
	M	S	1.000	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.750	0.500	1.000	1.000	1.000	1.000	1.000	0.750	1.000	1.000	1.000	1.000	0.500	1.000	1.000	0.500	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.750	1.000	1.000	0.500	1.000	0.500
	M	X	0.625	0.875	0.625	0.750	0.875	0.500	0.875	0.625	0.625	0.875	0.750	0.875	0.875	0.750	1.000	0.750	0.875	0.625	0.875	0.750	0.750	0.625	1.000	0.875	0.750	0.750	0.750	0.750	1.000	0.875	0.875	0.750	0.750	0.750	0.875	0.625	0.875	0.625	0.750	0.750	0.500	0.625	0.750	0.750	0.625	0.875	0.625	0.750	0.875	0.875
	M	Y	0.500	0.250	0.500	0.500	0.750	0.250	0.750	0.750	0.250	0.750	0.500	0.250	0.500	0.750	0.500	0.750	0.750	0.750	0.500	0.750	0.500	0.750	0.500	0.750	0.750	0.750	0.500	0.750	0.750	0.750	0.750	0.500	0.750	0.500	0.500	0.750	0.750	0.500	0.500	0.500	0.250	0.250	0.750	0.250	0.500	0.500	0.250	1.000	0.500	1.000

M：P 为 PPO，Q 为 QMIX，M 为 QMIX-MAT；K：D 为平均拦截距离，F 为弹药利用效率，R 为回合奖励，U 和 S 分别为敌方无人机、无人艇拦截率，X 和 Y 分别为我方无人机、无人艇损失率。

表 D.5: 第四章下层策略多场景测试原始数据

Cfg	M	K	0	1	2	3	4	5	6	7	8	9
A10S5	H	D	56.223	56.776	68.587	66.652	72.903	61.489	71.870	71.682	59.906	71.726
	H	W	0.000	0.500	0.333	0.500	0.600	0.500	0.571	0.625	0.667	0.700
	H	F	0.217	0.211	0.209	0.214	0.259	0.191	0.195	0.242	0.203	0.205
	R	D	58.581	56.092	57.178	52.009	58.604	55.631	54.771	61.157	53.581	54.268
	R	W	1.000	1.000	1.000	0.750	0.800	0.833	0.714	0.750	0.778	0.800
	R	F	0.234	0.263	0.227	0.246	0.263	0.227	0.265	0.273	0.254	0.221
A12S6	H	D	65.996	71.484	69.805	56.827	75.579	72.624	60.226	70.213	59.626	72.504
	H	W	1.000	1.000	0.667	0.500	0.600	0.667	0.571	0.625	0.667	0.600
	H	F	0.212	0.214	0.207	0.207	0.209	0.191	0.234	0.234	0.205	0.173
	R	D	55.214	58.632	52.201	56.520	57.220	61.015	62.626	59.769	59.953	56.459
	R	W	1.000	1.000	0.667	0.500	0.600	0.667	0.714	0.750	0.667	0.600
	R	F	0.243	0.243	0.246	0.230	0.240	0.228	0.234	0.243	0.239	0.195
A4S2	H	D	29.905	72.343	62.208	43.162	71.456	69.484	55.751	69.195	56.973	68.996
	H	W	0.000	0.500	0.333	0.250	0.400	0.333	0.429	0.500	0.444	0.500
	H	F	0.222	0.222	0.217	0.200	0.261	0.250	0.261	0.250	0.208	0.273
	R	D	46.834	45.192	44.566	41.071	27.953	55.458	44.033	46.379	49.294	42.892
	R	W	1.000	1.000	1.000	0.750	0.600	0.667	0.571	0.625	0.667	0.700
	R	F	0.261	0.250	0.222	0.227	0.267	0.261	0.250	0.240	0.273	0.231
A6S3	H	D	59.317	70.810	53.520	50.463	66.429	65.836	58.647	68.449	60.889	68.345
	H	W	1.000	0.500	0.333	0.250	0.400	0.500	0.571	0.625	0.667	0.700
	H	F	0.184	0.250	0.229	0.235	0.273	0.243	0.257	0.209	0.273	0.225
	R	D	49.106	45.368	50.094	51.376	47.408	48.148	54.313	45.461	55.573	49.986
	R	W	1.000	1.000	1.000	1.000	1.000	0.833	0.857	0.750	0.778	0.800
	R	F	0.257	0.225	0.281	0.250	0.257	0.211	0.257	0.200	0.257	0.257
A8S4	H	D	63.417	57.153	56.436	58.767	63.317	60.520	54.963	66.251	67.463	56.664
	H	W	1.000	1.000	0.667	0.750	0.800	0.833	0.714	0.750	0.778	0.800
	H	F	0.203	0.218	0.224	0.231	0.197	0.211	0.196	0.197	0.250	0.222
	R	D	48.399	54.407	52.363	56.272	46.387	55.572	47.797	58.843	60.365	45.924
	R	W	0.000	0.500	0.667	0.750	0.600	0.667	0.714	0.625	0.667	0.600
	R	F	0.237	0.245	0.286	0.250	0.213	0.226	0.255	0.234	0.261	0.222

Cfg	M	K	10	11	12	13	14	15	16	17	18	19
A10S5	H	D	63.109	68.074	64.167	59.471	68.094	71.082	55.878	64.419	69.324	66.256
	H	W	0.636	0.667	0.692	0.714	0.733	0.750	0.706	0.667	0.684	0.700
	H	F	0.194	0.185	0.211	0.254	0.211	0.221	0.232	0.187	0.195	0.217
	R	D	57.517	58.803	48.760	54.590	50.581	58.230	53.906	58.176	54.828	56.369
	R	W	0.818	0.833	0.769	0.786	0.733	0.750	0.765	0.778	0.737	0.750
	R	F	0.246	0.227	0.239	0.238	0.237	0.238	0.234	0.221	0.233	0.234
A12S6	H	D	50.357	71.296	67.393	72.937	66.711	61.815	64.691	65.447	69.535	70.650
	H	W	0.545	0.583	0.615	0.643	0.667	0.625	0.647	0.667	0.684	0.700
	H	F	0.188	0.220	0.200	0.220	0.214	0.188	0.250	0.214	0.243	0.184
	R	D	55.522	60.989	58.663	59.475	58.514	57.935	58.243	55.720	61.015	61.351
	R	W	0.636	0.667	0.692	0.714	0.733	0.750	0.765	0.722	0.737	0.750
	R	F	0.228	0.237	0.234	0.254	0.212	0.240	0.228	0.218	0.225	0.250
A4S2	H	D	38.599	43.156	61.911	81.875	50.312	68.646	34.848	70.621	66.204	63.976
	H	W	0.455	0.417	0.385	0.429	0.467	0.500	0.471	0.500	0.526	0.550
	H	F	0.235	0.217	0.227	0.240	0.273	0.261	0.188	0.222	0.273	0.261
	R	D	36.141	31.647	43.863	43.842	49.411	38.120	28.078	42.116	40.719	30.200
	R	W	0.727	0.667	0.692	0.714	0.733	0.688	0.647	0.667	0.684	0.650
	R	F	0.286	0.222	0.261	0.222	0.231	0.250	0.188	0.273	0.300	0.231
A6S3	H	D	71.946	68.922	67.650	49.111	61.539	60.072	60.587	63.422	60.530	55.798
	H	W	0.727	0.750	0.692	0.714	0.733	0.688	0.647	0.667	0.684	0.650
	H	F	0.225	0.231	0.250	0.200	0.225	0.281	0.233	0.273	0.214	0.212
	R	D	48.766	48.692	43.973	46.894	52.132	49.743	44.708	37.047	40.948	56.170
	R	W	0.818	0.833	0.769	0.786	0.800	0.750	0.765	0.722	0.737	0.750
	R	F	0.250	0.250	0.276	0.250	0.257	0.229	0.231	0.240	0.225	0.257
A8S4	H	D	67.181	63.871	65.865	77.152	70.210	63.145	68.253	59.074	59.187	70.497
	H	W	0.818	0.750	0.769	0.786	0.800	0.750	0.765	0.778	0.789	0.800
	H	F	0.214	0.212	0.255	0.226	0.235	0.196	0.255	0.222	0.207	0.203
	R	D	41.221	54.987	57.379	48.643	51.712	53.298	50.626	52.787	49.336	55.915
	R	W	0.545	0.583	0.615	0.643	0.667	0.688	0.706	0.722	0.737	0.750
	R	F	0.208	0.255	0.218	0.240	0.250	0.255	0.250	0.300	0.255	0.255

Cfg	M	K	20	21	22	23	24	25	26	27	28	29
A10S5	H	D	54.505	61.486	72.392	71.872	61.346	71.534	62.032	68.025	72.106	62.319
	H	W	0.714	0.682	0.696	0.708	0.680	0.692	0.704	0.714	0.724	0.733
	H	F	0.205	0.184	0.234	0.224	0.217	0.208	0.214	0.197	0.205	0.195
	R	D	59.604	53.103	54.979	51.851	57.502	55.587	51.255	58.340	54.289	51.138
	R	W	0.762	0.773	0.783	0.792	0.800	0.808	0.778	0.786	0.793	0.800
	R	F	0.246	0.242	0.246	0.254	0.259	0.221	0.231	0.242	0.221	0.214
A12S6	H	D	67.314	72.353	69.777	61.110	67.531	63.493	54.125	66.671	66.222	57.613
	H	W	0.714	0.727	0.739	0.750	0.760	0.731	0.704	0.714	0.690	0.667
	H	F	0.207	0.228	0.198	0.180	0.231	0.172	0.224	0.175	0.195	0.228
	R	D	57.935	63.825	65.634	53.571	62.424	56.702	62.827	55.758	61.547	57.839
	R	W	0.714	0.727	0.739	0.750	0.760	0.769	0.778	0.786	0.793	0.800
	R	F	0.230	0.240	0.250	0.217	0.212	0.265	0.217	0.243	0.225	0.243
A4S2	H	D	55.409	81.298	83.147	71.728	60.871	55.013	72.643	67.236	52.552	35.569
	H	W	0.524	0.545	0.565	0.542	0.520	0.538	0.556	0.571	0.586	0.567
	H	F	0.286	0.300	0.261	0.231	0.278	0.207	0.231	0.214	0.222	0.211
	R	D	39.475	50.223	61.079	47.232	48.677	38.598	42.662	33.985	31.717	37.184
	R	W	0.667	0.682	0.696	0.667	0.680	0.692	0.704	0.679	0.655	0.633
	R	F	0.214	0.300	0.300	0.263	0.273	0.273	0.261	0.188	0.250	0.192
A6S3	H	D	69.872	58.231	74.346	64.069	63.088	57.301	57.220	59.235	60.980	69.253
	H	W	0.667	0.682	0.696	0.708	0.720	0.692	0.704	0.679	0.690	0.700
	H	F	0.243	0.231	0.243	0.200	0.191	0.200	0.209	0.276	0.231	0.231
	R	D	50.443	55.030	52.920	47.139	49.049	43.551	54.810	45.678	43.328	48.557
	R	W	0.714	0.727	0.739	0.750	0.760	0.769	0.778	0.750	0.724	0.733
	R	F	0.267	0.237	0.250	0.257	0.237	0.250	0.257	0.205	0.222	0.243
A8S4	H	D	68.458	72.710	56.060	74.697	58.907	66.679	62.108	68.312	66.879	63.848
	H	W	0.810	0.818	0.783	0.792	0.760	0.769	0.741	0.750	0.759	0.733
	H	F	0.214	0.255	0.234	0.214	0.227	0.218	0.224	0.293	0.222	0.186
	R	D	62.283	47.750	55.646	60.356	56.459	40.703	49.071	52.486	47.869	54.716
	R	W	0.762	0.773	0.783	0.792	0.800	0.769	0.741	0.750	0.724	0.733
	R	F	0.255	0.231	0.267	0.279	0.240	0.216	0.239	0.250	0.250	0.245

Cfg	M	K	30	31	32	33	34	35	36	37	38	39
A10S5	H	D	58.836	71.929	62.540	66.481	74.008	68.392	60.532	65.644	66.023	68.816
	H	W	0.742	0.750	0.758	0.765	0.771	0.778	0.784	0.789	0.795	0.800
	H	F	0.234	0.200	0.211	0.238	0.188	0.192	0.214	0.203	0.221	0.234
	R	D	56.640	54.075	56.770	55.534	57.281	52.279	56.723	61.449	53.671	53.944
	R	W	0.806	0.812	0.818	0.824	0.829	0.833	0.838	0.842	0.846	0.825
	R	F	0.259	0.224	0.254	0.224	0.278	0.246	0.224	0.246	0.254	0.237
A12S6	H	D	75.931	73.220	64.566	65.285	72.437	66.038	68.137	70.974	61.740	58.100
	H	W	0.677	0.688	0.667	0.676	0.686	0.694	0.703	0.711	0.692	0.675
	H	F	0.247	0.196	0.210	0.200	0.209	0.205	0.222	0.209	0.179	0.197
	R	D	58.135	54.007	59.155	61.461	60.523	57.929	61.586	61.832	53.514	54.938
	R	W	0.774	0.781	0.788	0.794	0.800	0.806	0.811	0.816	0.821	0.800
	R	F	0.250	0.243	0.217	0.257	0.261	0.214	0.286	0.257	0.240	0.195
A4S2	H	D	73.593	43.014	34.778	70.248	54.986	32.690	68.122	57.230	59.322	80.652
	H	W	0.581	0.562	0.545	0.559	0.571	0.556	0.541	0.553	0.564	0.575
	H	F	0.250	0.200	0.192	0.194	0.240	0.130	0.240	0.250	0.261	0.240
	R	D	33.117	34.678	37.306	34.484	40.368	35.609	49.274	45.105	37.886	35.511
	R	W	0.613	0.594	0.576	0.559	0.571	0.583	0.595	0.605	0.590	0.600
	R	F	0.192	0.227	0.217	0.238	0.222	0.250	0.263	0.261	0.263	0.261
A6S3	H	D	75.716	59.517	58.054	47.040	59.282	63.952	62.360	59.644	60.945	67.516
	H	W	0.710	0.719	0.697	0.706	0.714	0.722	0.703	0.711	0.692	0.700
	H	F	0.243	0.243	0.222	0.231	0.237	0.225	0.229	0.243	0.250	0.220
	R	D	46.813	38.067	51.397	50.043	48.175	44.385	39.564	44.369	45.506	57.312
	R	W	0.742	0.719	0.727	0.706	0.714	0.694	0.676	0.684	0.667	0.675
	R	F	0.237	0.263	0.214	0.259	0.205	0.250	0.259	0.243	0.242	0.273
A8S4	H	D	68.364	57.583	72.665	69.236	60.482	56.487	55.770	74.137	64.740	65.924
	H	W	0.742	0.719	0.727	0.735	0.743	0.750	0.757	0.763	0.769	0.775
	H	F	0.235	0.192	0.240	0.200	0.231	0.214	0.214	0.240	0.231	0.245
	R	D	55.692	50.609	52.000	55.827	47.804	55.139	59.244	53.226	50.009	55.957
	R	W	0.742	0.750	0.758	0.765	0.743	0.750	0.757	0.763	0.744	0.750
	R	F	0.235	0.231	0.250	0.267	0.229	0.255	0.240	0.235	0.233	0.255

Cfg	M	K	40	41	42	43	44	45	46	47	48	49
A10S5	H	D	70.964	57.908	69.080	61.084	66.989	60.929	64.311	64.667	74.902	68.445
	H	W	0.805	0.810	0.814	0.818	0.822	0.826	0.830	0.833	0.837	0.840
	H	F	0.208	0.195	0.214	0.234	0.234	0.242	0.208	0.259	0.273	0.179
	R	D	53.193	54.246	53.530	54.214	56.316	56.363	53.281	59.918	51.883	59.067
	R	W	0.829	0.810	0.791	0.795	0.800	0.804	0.809	0.812	0.796	0.780
	R	F	0.227	0.220	0.206	0.208	0.234	0.214	0.238	0.246	0.237	0.259
A12S6	H	D	68.400	74.120	65.001	64.721	56.880	74.069	67.869	63.847	61.164	61.904
	H	W	0.683	0.690	0.698	0.705	0.689	0.696	0.702	0.708	0.694	0.700
	H	F	0.228	0.220	0.194	0.207	0.185	0.237	0.222	0.191	0.178	0.212
	R	D	57.486	56.195	55.525	62.291	59.040	52.895	60.795	57.213	63.214	58.353
	R	W	0.805	0.810	0.791	0.795	0.800	0.783	0.787	0.792	0.796	0.800
	R	F	0.234	0.220	0.215	0.250	0.243	0.225	0.240	0.222	0.237	0.222
A4S2	H	D	57.504	61.289	64.324	50.053	57.220	65.588	48.961	77.181	48.437	56.805
	H	W	0.585	0.571	0.558	0.545	0.533	0.543	0.532	0.542	0.531	0.540
	H	F	0.250	0.156	0.261	0.238	0.200	0.261	0.176	0.261	0.267	0.261
	R	D	29.510	44.800	36.927	45.230	32.806	34.119	49.610	43.368	46.482	32.995
	R	W	0.585	0.595	0.581	0.591	0.600	0.587	0.596	0.604	0.612	0.620
	R	F	0.217	0.261	0.273	0.240	0.222	0.235	0.222	0.207	0.240	0.240
A6S3	H	D	65.861	65.904	67.441	70.255	72.679	59.239	65.510	58.310	48.770	73.960
	H	W	0.707	0.714	0.721	0.727	0.733	0.739	0.745	0.750	0.755	0.760
	H	F	0.231	0.273	0.250	0.257	0.214	0.220	0.257	0.188	0.191	0.225
	R	D	40.914	50.806	47.504	34.764	45.492	46.585	53.899	53.412	48.014	53.644
	R	W	0.683	0.690	0.698	0.682	0.689	0.674	0.681	0.688	0.673	0.680
	R	F	0.225	0.281	0.265	0.200	0.290	0.296	0.300	0.250	0.242	0.243
A8S4	H	D	59.994	66.276	68.655	50.415	67.457	57.955	65.644	69.527	64.754	57.146
	H	W	0.780	0.786	0.791	0.773	0.778	0.761	0.766	0.771	0.776	0.760
	H	F	0.250	0.214	0.214	0.182	0.214	0.224	0.222	0.226	0.235	0.216
	R	D	52.636	59.362	48.425	50.962	56.311	51.197	46.489	44.872	45.010	62.880
	R	W	0.732	0.738	0.744	0.727	0.733	0.739	0.745	0.729	0.735	0.740
	R	F	0.204	0.255	0.222	0.256	0.222	0.245	0.261	0.208	0.267	0.250

M：H 为 HGAT-PPO，R 为 Rule；K：D 为平均拦截距离，W 为任务胜率，F 为弹药利用效率。

表 D.6: 第五章上层决策对比实验原始数据

M	K	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
F	W	1.000	0.000	1.000	0.000	1.000	0.000	1.000	1.000	0.000	1.000	0.000	0.000	0.000	1.000	1.000	1.000	0.000	1.000	1.000	0.000	1.000	1.000	1.000	1.000	1.000	0.000	1.000	1.000	0.000	1.000
F	I	1.000	0.281	1.000	0.875	1.000	0.906	1.000	1.000	0.375	1.000	0.250	0.375	0.344	1.000	1.000	1.000	0.969	1.000	1.000	0.938	1.000	1.000	1.000	1.000	1.000	0.906	1.000	1.000	0.938	1.000
F	L	0.656	0.188	0.688	0.781	0.719	0.844	0.594	0.656	0.438	0.656	0.344	0.344	0.250	0.781	0.781	0.750	0.750	0.781	0.656	0.938	0.750	0.688	0.531	0.812	0.750	0.688	0.750	0.719	0.750	0.750
F	D	56.119	45.245	53.627	50.803	53.775	44.341	58.970	55.871	43.575	49.794	51.286	49.096	45.988	48.712	50.163	54.536	51.275	41.899	48.870	44.533	52.229	53.505	60.007	47.696	53.157	52.714	51.340	48.475	52.049	52.851
L	W	1.000	1.000	0.000	0.000	0.000	1.000	0.000	0.000	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	0.000	0.000	0.000	1.000	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	0.000	0.000
L	I	1.000	1.000	0.469	0.438	0.438	1.000	0.344	0.438	1.000	0.344	1.000	1.000	1.000	1.000	1.000	1.000	0.406	0.312	0.312	1.000	1.000	0.469	1.000	1.000	1.000	1.000	1.000	1.000	0.469	0.438
L	L	0.656	0.531	0.344	0.219	0.219	0.594	0.188	0.250	0.500	0.281	0.375	0.656	0.594	0.656	0.531	0.656	0.375	0.281	0.375	0.781	0.531	0.188	0.719	0.594	0.531	0.562	0.688	0.719	0.406	0.312
L	D	75.975	63.369	87.772	78.324	86.626	69.692	76.484	83.025	84.545	70.905	75.121	65.420	72.207	60.645	66.819	67.038	59.245	68.939	79.534	70.792	68.572	98.250	66.714	74.242	71.988	74.927	69.469	74.840	65.252	87.085

M：F 为固定规则策略，L 为大模型辅助决策；K：W 为任务胜率，I 为目标拦截率，L 为平台损失率，D 为平均拦截距离。